A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Impulse & 2D Collisions

At AS you saw that momentum is conserved when things collide in a straight line. This topic finishes the story. First we look at what a force does to momentum over time, which is called **impulse** and explains crumple zones, airbags and why you bend your knees when you land. Then we take collisions off the line and onto a table top, where balls fly off at angles, and learn how to tell whether any kinetic energy was lost on the way.

Momentum and Newton's second law

Quick recap from Forces & Momentum: the momentum of an object is its mass times its velocity. Because velocity is a vector, **momentum is a vector** too. It points the same way as the velocity, and in a calculation anything moving the "wrong" way gets a minus sign.

Momentum

FORMULA SHEET

$$p = mv$$

p: momentum / kg m s^{-1} m: mass / kg v: velocity / m s^{-1}

You already know $F = ma$. Newton actually wrote his second law in terms of momentum, and that version turns out to be the more useful one for impacts and collisions. Let's see that the two say the same thing.

From F = ma to F = Δp/Δt

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Start with Newton's second law for a constant resultant force F acting on a mass m:

$$F = ma$$

2. The force is constant, so the acceleration is constant. It is the change in velocity divided by the time taken, where u is the velocity before and v the velocity after:

$$F = m\,\frac{v - u}{\Delta t}$$

3 Multiply out the bracket. The mass is constant, so it can go inside:

$$F = \frac{mv - mu}{\Delta t}$$

mv is the final momentum and mu the initial momentum, so the top line is the change in momentum Δp.

4 So the resultant force equals the rate of change of momentum:

$$\boxed{F = \frac{\Delta p}{\Delta t}}$$

We assumed the mass stays the same. $F = \Delta p / \Delta t$ still works when it doesn't, which is why Newton's own version is the more general one.

DEFINITION

Newton's second law (momentum form)

The **resultant force** acting on an object is equal to the **rate of change of its momentum**, and acts in the direction of the change in momentum.

Impulse

Rearrange $F = \Delta p / \Delta t$ and you get the most useful equation in this topic. The quantity $F\Delta t$ (force multiplied by the time it acts for) is called the **impulse**, and it equals the change in momentum it causes.

Impulse

FORMULA SHEET

$$F\Delta t = \Delta p$$

F: resultant force / N Δt: time the force acts / s Δp: change in momentum / kg m s^{-1} or N s

DEFINITION

Impulse

The impulse of a force is the product of the force and the time for which it acts, $F\Delta t$. It is equal to the **change in momentum** of the object it acts on. Impulse is a vector, measured in **N s** (which is the same as

kg m s⁻¹).

In a real impact the force is not constant: it rises as the objects squash together and falls as they separate. Each thin strip under a force–time graph has area $F\,\delta t$, which is a little bit of impulse, so adding them all up gives the key result:

KEY IDEA

Impulse is the area under a force–time graph

Area under the F–t graph = impulse = change in momentum. For a constant force the area is a rectangle, $F\Delta t$. For a varying force, count squares or split the area into triangles and rectangles. The average force is the impulse divided by the contact time.

Same change in momentum, longer time, smaller force

Suppose you need to stop something. Its change in momentum is fixed (it goes from mv to zero), so the impulse needed is fixed. But impulse is force *times* time, so if you can make the stop take ten times longer, the average force is ten times smaller. That one idea explains a lot of safety design:

- **Crumple zones** fold up over about a tenth of a second instead of the car stopping almost instantly against a wall.
- **Airbags and seatbelts** stretch or squash, so your head and chest take longer to stop.
- **Cycle helmets, crash mats and trainers** have soft foam that increases the stopping time.
- **Bending your knees** when you land, or "giving" with your hands when you catch a ball, does the same job with your own body.

Sport uses the idea the other way round. When you **follow through** with a racket, bat or kick, the force acts on the ball for longer, so for the same force the ball gets a bigger impulse and leaves faster.

WORKED EXAMPLE

A tennis ball of mass 0.058 kg hits a wall at 25 m s⁻¹ and rebounds along the same line at 20 m s⁻¹. It is in contact with the wall for 4.0 ms. Calculate (a) the impulse on the ball and (b) the average force the wall exerts on it.

Show solution

(a) Choose a positive direction first: **towards the wall is positive**. Then $u = +25\ \text{m s}^{-1}$ and $v = -20\ \text{m s}^{-1}$.

$$\Delta p = m(v - u) = 0.058 \times (-20 - 25) = 0.058 \times (-45) = -2.61\ \text{N s}$$

The impulse is $2.6\ \text{N s}$ directed away from the wall.

(b)

$$F = \frac{\Delta p}{\Delta t} = \frac{-2.61}{4.0 \times 10^{-3}} = -653\ \text{N}$$

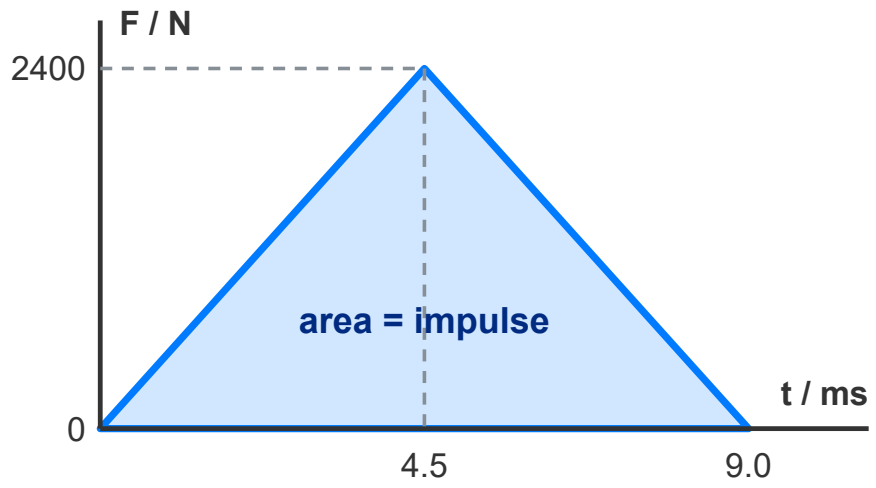
So the average force is about $650\ \text{N}$ away from the wall. That is over a thousand times the ball's weight, which is why the ball squashes visibly.

COMMON MISTAKE

When something bounces back, its velocity changes sign. Writing $\Delta p = 0.058 \times (20 - 25)$ gives $-0.29\ \text{N s}$, which is wrong by a factor of about nine. A rebound always gives a **bigger** change in momentum than just stopping, because the object has to be stopped and then sent back.

WORKED EXAMPLE

A football of mass 0.45 kg is at rest. The graph shows how the force from a boot varies during the kick. Find the speed of the ball as it leaves the boot, and the average force during the kick.



Force on the ball during the kick.

Show solution

The impulse is the area of the triangle. Convert the time to seconds first:

$$F\Delta t = \tfrac{1}{2} \times 9.0 \times 10^{-3} \times 2400 = 10.8 \text{ N s}$$

The ball starts at rest, so all of this is its final momentum:

$$v = \frac{\Delta p}{m} = \frac{10.8}{0.45} = 24 \text{ m s}^{-1}$$

The average force is the impulse spread over the whole contact time:

$$F_{\text{avg}} = \frac{10.8}{9.0 \times 10^{-3}} = 1200 \text{ N}$$

For a triangle the average force is exactly half the peak, which is a nice check.

Core Practical 9: force and change of momentum

CORE PRACTICAL

Investigate the relationship between the force exerted on an object and its change of momentum

A trolley runs along a horizontal runway, pulled by a string that passes over a pulley at the end of the bench to a hanging mass. A **force sensor** on the trolley measures the tension F in the string. Two **light gates**, connected to a data logger, time a card of known length L on top of the trolley, giving its speeds $u = L/t_1$ and $v = L/t_2$ at the two gates, and the time Δt taken to travel between them.

1. **Compensate for friction**: raise the start end of the runway slightly until a gently pushed trolley passes through both gates at the same speed (equal card times).
2. Measure the mass m of the trolley, including the sensor and card, with a balance. Measure L with vernier calipers.
3. Release the trolley from rest. Record F, u, v and Δt. Repeat with different hanging masses, and repeat each one to get a mean.
4. For each run calculate the impulse $F\Delta t$ and the change in momentum $\Delta p = m(v - u)$.

What to plot: Δp on the y-axis against $F\Delta t$ on the x-axis. If $F\Delta t = \Delta p$, the points lie on a **straight line through the origin with a gradient of 1**. (Another version keeps F fixed and moves the second gate to change Δt. Then Δp against Δt is a straight line through the origin whose gradient is F.)

- If friction is not compensated, the resultant force is less than the tension, so every Δp comes out smaller than $F\Delta t$. The points fall below the line of gradient 1 and the best-fit line no longer passes through the origin. That is a **systematic** error.
- Keep the string parallel to the runway, so all of the tension pulls along the runway.
- Use a long card and a data logger: hand timing over these short times would swamp the result with reaction-time error.

- **Safety:** put a box with padding under the hanging masses so they don't land on feet, and a buffer at the end of the runway to stop the trolley.

Conservation of momentum in two dimensions

DEFINITION

Principle of conservation of linear momentum

In a **closed system** (one with no external resultant force acting), the **total momentum stays constant**. In a collision or explosion, the total momentum before equals the total momentum after.

Why is this true? During a collision, A pushes on B and B pushes back on A with an equal and opposite force (Newton's third law) for exactly the same time. So the impulses on A and B are equal and opposite: whatever momentum A loses, B gains. That argument works in any direction, which is the key to 2D problems.

On a snooker table or an air-hockey table, objects fly off at angles. Momentum is a vector, so we deal with it in one of two ways:

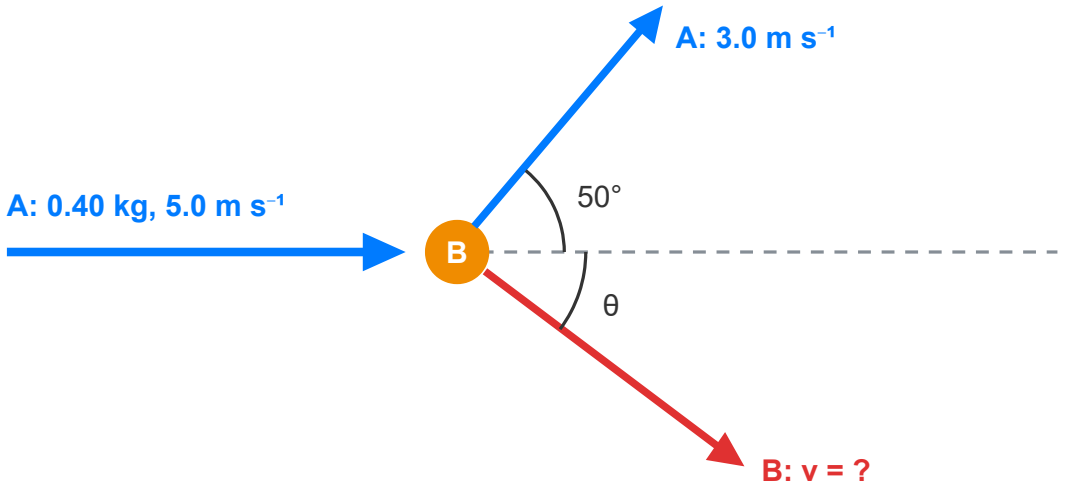
- **Resolve into components.** Pick x and y directions at right angles. Momentum is conserved separately in each direction: total p_x before = total p_x after, and total p_y before = total p_y after. That gives two equations, so you can find two unknowns (often a speed and an angle).
- **Draw a vector diagram.** Draw the momentum vectors after the collision tip-to-tail. Their resultant must be the same arrow as the total momentum before.

EXAM TIP

Lay the x-axis along the direction of the incoming object. Then the total p_y before is zero, so the y-components after must cancel: if one object goes off above the line, the other **must** go off below it.

WORKED EXAMPLE

Puck A (mass 0.40 kg) slides at 5.0 m s^{-1} and hits puck B (mass 0.60 kg), which is at rest. Afterwards A moves at 3.0 m s^{-1} at 50° to its original direction. Find (a) the speed and direction of B and (b) whether the collision is elastic.



Take x along A's original direction and y upwards.

Show solution

x-direction. Before: $0.40 \times 5.0 = 2.00\ \text{kg m s}^{-1}$. After, A contributes $0.40 \times 3.0\cos 50^\circ = 0.771\ \text{kg m s}^{-1}$, so

$$0.60\,v_{Bx} = 2.00 - 0.771 = 1.229 \quad \Rightarrow \quad v_{Bx} = 2.05\ \text{m s}^{-1}$$

y-direction. Before: zero. After, A contributes $+0.40 \times 3.0\sin 50^\circ = +0.919\ \text{kg m s}^{-1}$, so B must have -0.919:

$$0.60\,v_{By} = -0.919 \quad \Rightarrow \quad v_{By} = -1.53\ \text{m s}^{-1}$$

Combine the components with Pythagoras and tan:

$$v_B = \sqrt{2.05^2 + 1.53^2} = 2.56\ \text{m s}^{-1}, \qquad \theta = \tan^{-1}\left(\frac{1.53}{2.05}\right) = 36.8^\circ$$

So B moves at $2.6\ \text{m s}^{-1}$ at 37° below A's original direction.

(b) Kinetic energy is a scalar, so just add the values for each object, using the full speeds:

$$E_{\text{before}} = \tfrac{1}{2}(0.40)(5.0)^2 = 5.0\ \text{J}$$

$$E_{\text{after}} = \tfrac{1}{2}(0.40)(3.0)^2 + \tfrac{1}{2}(0.60)(2.56)^2 = 1.80 + 1.96 = 3.8\ \text{J}$$

Kinetic energy has fallen by about 1.2 J (a quarter of it), so the collision is **inelastic**.

Explosions in two dimensions

An explosion is a collision run backwards. If the object starts at rest, the total momentum before is zero, so the momenta of all the pieces afterwards must add up to zero. With two pieces, they fly apart in exactly opposite directions with equal-sized momenta (that's recoil). With three pieces, the three momentum vectors drawn tip-to-tail must form a **closed triangle**.

Elastic and inelastic collisions

In a closed system **momentum is always conserved**. Kinetic energy is a different story. Total energy is always conserved too, but some of the kinetic energy can be transferred to other stores during the impact: the objects warm up slightly, they make a sound, and they may be permanently bent or dented.

DEFINITION

Elastic and inelastic collisions

In an **elastic** collision the total kinetic energy of the objects is the same before and after. In an **inelastic** collision the total kinetic energy after is less than before (momentum is still conserved).

How to decide

1. Use momentum conservation (in each direction) to find any unknown velocities.
2. Find the **speed** of every object before and after. In 2D, combine the components first with Pythagoras.
3. Work out $\frac{1}{2}mv^2$ (or $p^2/2m$) for every object and add them up, before and after.
4. Equal totals: elastic. Smaller total after: inelastic. If the objects stick together, it is certainly inelastic. A **bigger** total after means energy came from somewhere else, as in an explosion, where chemical or elastic potential energy is released.

COMMON MISTAKE

Kinetic energy is a **scalar**. Never resolve it into components or give it a minus sign for moving left. A ball moving left at 3 m s^{-1} has exactly the same kinetic energy as one moving right at 3 m s^{-1}, even though their momenta have opposite signs.

Collisions between hard steel ball bearings or snooker balls are very nearly elastic, and in kinetic theory collisions between gas molecules are taken to be perfectly elastic. A car crash, a ball of Plasticine hitting a wall, or a dart sticking into a dartboard are strongly inelastic.

Kinetic energy in terms of momentum

Collision problems are often easier if you can jump straight from momentum to kinetic energy without finding the velocity first. The link is a two-line derivation you need to be able to reproduce.

Deriving $E_k = p^2/2m$ SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 Start from the kinetic energy of a mass m moving at speed v:

$$E_k = \tfrac{1}{2}mv^2$$

2 Multiply the top and bottom by m. This changes nothing (it's multiplying by 1) but builds an mv on top:

$$E_k = \frac{m \cdot mv^2}{2m} = \frac{m^2v^2}{2m} = \frac{(mv)^2}{2m}$$

3 The bracket on top is the momentum, $p = mv$:

$$\boxed{E_k = \frac{p^2}{2m}}$$

Rearranged, $p = \sqrt{2mE_k}$. Use whichever direction the question needs.

Kinetic energy of a non-relativistic particle

FORMULA SHEET

$$E_k = \frac{p^2}{2m}$$

E_k: kinetic energy / J p: momentum / kg m s^{-1} m: mass / kg

"Non-relativistic" means much slower than light. Both $p = mv$ and $E_k = \frac{1}{2}mv^2$ start to go wrong as speeds approach 3.00×10^8 m s^{-1}, so this equation does too. For electrons accelerated through a few thousand volts it is fine.

KEY IDEA

Same momentum, different energy

When something at rest explodes into two pieces, they get **equal** sized momenta p. Then $E_k = p^2/2m$ says the lighter piece gets more kinetic energy, in the inverse ratio of the masses: $E_1/E_2 = m_2/m_1$. That's why a bullet carries almost all the energy and the gun's recoil is gentle, and why an alpha particle carries nearly all the energy released when a nucleus decays.

WORKED EXAMPLE

A stationary nucleus emits an alpha particle of mass 6.64×10^{-27} kg with kinetic energy 7.7×10^{-13} J. The nucleus left behind has mass 3.69×10^{-25} kg. Calculate its recoil kinetic energy and the fraction of the total energy carried by the alpha particle.

Show solution

Momentum of the alpha particle:

$$p = \sqrt{2mE_k} = \sqrt{2 \times 6.64 \times 10^{-27} \times 7.7 \times 10^{-13}} = 1.01 \times 10^{-19} \text{ kg m s}^{-1}$$

The total momentum was zero, so the nucleus recoils with the same size momentum in the opposite direction:

$$E_k = \frac{p^2}{2m} = \frac{(1.01 \times 10^{-19})^2}{2 \times 3.69 \times 10^{-25}} = 1.4 \times 10^{-14} \text{ J}$$

Fraction carried by the alpha particle:

$$\frac{7.7 \times 10^{-13}}{7.7 \times 10^{-13} + 1.4 \times 10^{-14}} = 0.98$$

Check with the mass ratio: $E_{\text{nucleus}}/E_\alpha = m_\alpha/m_{\text{nucleus}} = 0.018$, and $0.018 \times 7.7 \times 10^{-13} = 1.4 \times 10^{-14}$ J.

The same equation links to the de Broglie wavelength from Unit 2. An electron accelerated through 150 V gains $E_k = eV = 1.60 \times 10^{-19} \times 150 = 2.40 \times 10^{-17}$ J. Its momentum is $p = \sqrt{2mE_k} = \sqrt{2 \times 9.11 \times 10^{-31} \times 2.40 \times 10^{-17}} = 6.61 \times 10^{-24}$ kg m s^{-1}, so $\lambda = h/p = 6.63 \times 10^{-34}/6.61 \times 10^{-24} = 1.0 \times 10^{-10}$ m: about the spacing of atoms in a crystal, which is why electron beams diffract.

Core Practical 10: analysing collisions with ICT

CORE PRACTICAL

Use ICT to analyse collisions between small spheres, e.g. ball bearings on a table top

Ball A is rolled down a short ramp so that it strikes ball B, which is at rest on a flat, level table, slightly off-centre. A camera (a phone on a stand is fine) films the collision from **directly above**, with a metre rule or grid in the same plane as the balls for scale.

1. Measure the mass of each ball with a top-pan balance.
2. Load the video into analysis software (such as Tracker). Set the scale using the ruler, set the origin and axes, and mark the centre of each ball frame by frame.
3. The software plots x and y against t for each ball. The **gradient** of each straight section gives v_x and v_y before and after the collision.
4. Calculate the total p_x and total p_y before and after. They should agree within experimental uncertainty.
5. Calculate the total kinetic energy before and after to decide whether the collision was elastic.

- **Camera not directly overhead** distorts distances (parallax). Use a spirit level on the camera and keep the scale rule in the plane of the balls.
- **Motion blur** makes centres hard to mark. Use bright lighting and a high frame rate.
- Friction slowly slows the balls, so use frames close to the collision, and fit a line through several points rather than using just two.
- Rolling balls also have rotational kinetic energy, so the "missing" energy you calculate is not all lost in the impact.

Quick checks

QUICK CHECK

A 0.50 kg ball hits a wall at 4.0 m s^{-1} and bounces straight back at 4.0 m s^{-1}. What is the magnitude of its change in momentum?

- 0 N s
- 2.0 N s
- 4.0 N s
- 8.0 N s

The speed is unchanged but the velocity reverses: $\Delta p = 0.50 \times (-4.0 - 4.0) = -4.0$ N s. Zero is the trap: momentum is a vector, so a change of direction is a change of momentum.

QUICK CHECK

Why does an airbag reduce the force on a driver's head in a crash?

- It reduces the head's change in momentum.
- It increases the time taken for the head to stop, so a smaller force gives the same impulse.
- It absorbs the head's momentum so momentum is not conserved.
- It reduces the impulse on the head.

The head goes from moving to stopped either way, so Δp (and the impulse) is the same. $F = \Delta p / \Delta t$, so a longer Δt means a smaller average force.

QUICK CHECK

Puck A moves along the x-axis and hits stationary puck B. Afterwards B moves off at 30° below the x-axis. Which statement about A after the collision must be true?

- A moves along the x-axis.
- A has a component of velocity above the x-axis (or A is at rest).
- A moves at 30° below the x-axis.
- A must stop.

Before the collision the total y-momentum is zero, so afterwards A's y-momentum must cancel B's downward component. The only way A could avoid this is by having no momentum at all.

QUICK CHECK

Particles X and Y have the same momentum. X has four times the mass of Y. What is the ratio (kinetic energy of X) : (kinetic energy of Y)?

- 1 : 4
- 4 : 1
- 1 : 2
- 1 : 16

$E_k = p^2/2m$. With the same p, kinetic energy is inversely proportional to mass, so the heavier particle has a quarter of the energy.

QUICK CHECK

Two trolleys collide and stick together. No external forces act. Which statement is correct?

- Total kinetic energy and total momentum are both conserved.
- Total kinetic energy is conserved but total momentum is not.
- Total momentum is conserved but total kinetic energy is not.
- Neither is conserved, but total energy is.

With no external force, momentum is always conserved. Objects sticking together is always inelastic, so some kinetic energy is transferred to thermal energy and sound. Total energy is conserved too, which is why the last option is a trap: momentum is conserved as well.

QUICK CHECK

Which of these is not a correct unit for impulse?

- N s
- kg m s^{-1}
- N s^{-1}
- J s m^{-1}

Impulse is force × time, so N s, which equals kg m s^{-2} × s = kg m s^{-1}. Since 1 J = 1 N m, J s m^{-1} is also N s. N s^{-1} would be a rate of change of force.

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Circular Motion

Anything moving in a circle is accelerating, even if its speed never changes, because its direction is changing all the time. That acceleration points at the centre of the circle, and it needs a resultant force pointing at the centre too. In this topic we measure angles in radians, describe spinning with angular velocity, derive the centripetal acceleration from a vector diagram, and then use $F = mv^2/r$ for cars on bends, balls on strings and roller coasters.

Measuring angles in radians

Degrees are a human choice: someone decided a full turn should be 360. Physicists prefer an angle unit that comes from the circle itself. Take a circle of radius r, walk a distance s round the edge, and the angle you have turned through at the centre is

$$\theta = \frac{s}{r} \qquad (\text{in radians})$$

DEFINITION

The radian

One **radian** is the angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle.

So an angle of 1 rad means "the arc is one radius long". A full turn has an arc of one circumference, $2\pi r$, so a full turn is $2\pi r/r = 2\pi$ radians. That gives the conversion:

$$2\pi \text{ rad} = 360^\circ \qquad \Rightarrow \qquad 1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57.3^\circ$$

- **Degrees to radians:** multiply by $\frac{\pi}{180}$.
- **Radians to degrees:** multiply by $\frac{180}{\pi}$.

Degrees	30°	45°	60°	90°	180°	270°	360°
Radians (exact)	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$	2π
Radians (decimal)	0.524	0.785	1.05	1.57	3.14	4.71	6.28

EXAM TIP

Check your calculator mode. In circular motion questions, use **radians** for anything involving ω and $\theta = s/r$.

Because the radian is a ratio of two lengths it has no real dimensions, which is why ω in rad s^{-1} can be written as just s^{-1} in a units check.

Angular velocity

For something going round, "how fast" can mean two things. A horse near the edge of a carousel covers more metres per second than one near the middle, but both go round once in the same time. **Angular velocity** captures the second idea: how quickly the angle changes.

DEFINITION

Angular velocity

The **angular velocity** ω of an object moving in a circle is the angle swept out by the radius per unit time, $\omega = \dfrac{\Delta\theta}{\Delta t}$. It is measured in rad s^{-1}.

In one complete revolution the angle is 2π and the time taken is the period T, so $\omega = 2\pi/T$. Rearranged, this is the formula-sheet version. Since frequency $f = 1/T$, you can also write $\omega = 2\pi f$.

Period

FORMULA SHEET

$$T = \frac{2\pi}{\omega}$$

T: time for one revolution / s

ω: angular velocity / rad s^{-1}

Linear speed

FORMULA SHEET

$$v = \omega r$$

v: speed along the circle / m s^{-1} r: radius / m

Where v = ωr comes from

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. In a short time Δt the object turns through an angle $\Delta\theta$. By the definition of angular velocity:

$$\Delta\theta = \omega\,\Delta t$$

2. The distance travelled along the circle is the arc length. From the definition of the radian, $s = r\theta$:

$$\Delta s = r\,\Delta\theta = r\,\omega\,\Delta t$$

3 Speed is distance divided by time:

$$v = \frac{\Delta s}{\Delta t} = \frac{r\,\omega\,\Delta t}{\Delta t} \quad \Rightarrow \quad \boxed{v = \omega r}$$

This only works with ω in radians per second, because $s = r\theta$ is only true in radians.

KEY IDEA

Same ω, different v

Every point on a rigid spinning object (a wheel, a disc, the Earth) has the **same angular velocity**. Its speed grows in proportion to its distance from the axis, $v = \omega r$. Twice as far out means twice as fast.

WORKED EXAMPLE

The Earth turns once every 24 hours. Its radius is 6.37×10^6 m. Calculate (a) the Earth's angular velocity, (b) the speed of a person on the equator, and (c) the speed of a person in London, at latitude 51.5°.

Show solution

(a) Convert the period to seconds: $T = 24 \times 3600 = 86\,400$ s.

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{86\,400} = 7.27 \times 10^{-5} \text{ rad s}^{-1}$$

(b) On the equator the radius of the circle is the Earth's radius:

$$v = \omega r = 7.27 \times 10^{-5} \times 6.37 \times 10^6 = 463 \text{ m s}^{-1}$$

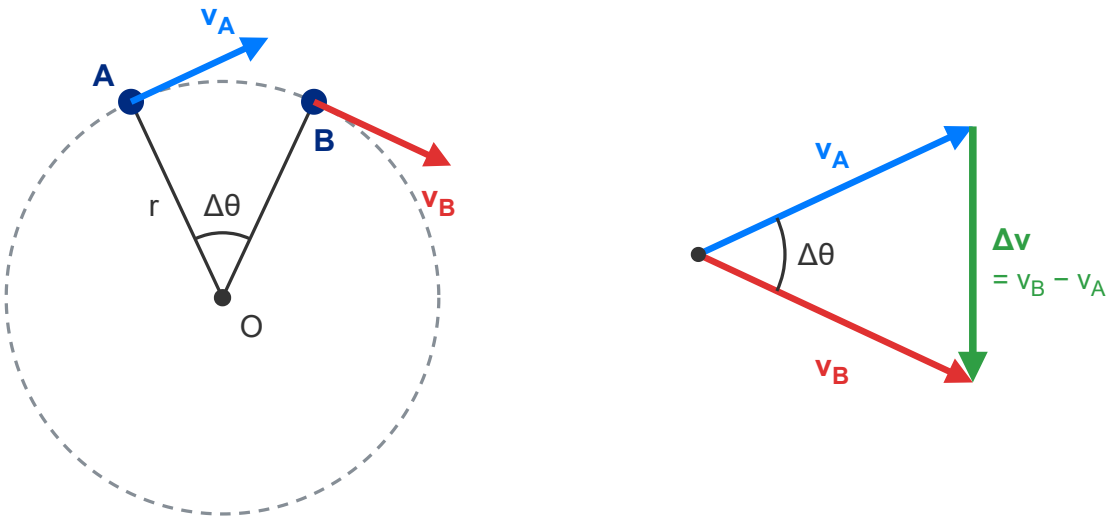
(c) In London you travel round a smaller circle, whose radius is the distance from the Earth's axis: $r = R\cos 51.5^\circ = 6.37 \times 10^6 \times 0.623 = 3.97 \times 10^6$ m. The angular velocity is the same:

$$v = 7.27 \times 10^{-5} \times 3.97 \times 10^6 = 288 \text{ m s}^{-1}$$

Both of you go round once a day (same ω), but the person on the equator has further to go.

Centripetal acceleration

Velocity is a vector. Even when the speed is constant, the **direction** of the velocity changes as an object goes round a circle, so the velocity changes, so the object is accelerating. Which way? The best way to find out is to draw the velocity at two nearby moments and look at the difference.



Left: the object moves from A to B at constant speed. Right: the same two velocity vectors drawn from one point. Δv joins the tip of v_A to the tip of v_B, and it points straight down the page, towards the centre of the circle.

Deriving a = v²/r with a vector diagram

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 An object moves round a circle of radius r at constant speed v. In a short time Δt it moves from A to B, and the radius turns through a small angle $\Delta\theta$. The velocity is always at 90° to the radius, so the velocity also turns through $\Delta\theta$:

$$|v_A| = |v_B| = v, \qquad \text{angle between } v_A \text{ and } v_B = \Delta\theta$$

2 The change in velocity is $\Delta v = v_B - v_A$. Draw v_A and v_B tail to tail: Δv is the arrow from the tip of v_A to the tip of v_B. This is an isosceles triangle with two sides of length v and the angle $\Delta\theta$ between them.

By symmetry, Δv is perpendicular to the line halfway between v_A and v_B, so it points along the radius at the midpoint of the arc, towards O.

3 When $\Delta\theta$ is small, the short side of the triangle is almost the same as an arc of a circle of radius v. Arc length = radius × angle, so

$$|\Delta v| \approx v\,\Delta\theta$$

4 Acceleration is change in velocity over time:

$$a = \frac{|\Delta v|}{\Delta t} = v\,\frac{\Delta\theta}{\Delta t} = v\,\omega$$

$\Delta\theta/\Delta t$ is the angular velocity ω.

5 From $v = \omega r$, $\omega = v/r$. Substitute:

$$a = v \times \frac{v}{r} \quad \Rightarrow \quad \boxed{a = \frac{v^2}{r}}$$

As $\Delta t \to 0$, A and B merge, the approximation in step 3 becomes exact, and Δv points exactly at the centre. The acceleration is directed **towards the centre** of the circle: it is **centripetal** ("centre-seeking").

Deriving a = rω² SPEC: YOU MUST BE ABLE TO DERIVE THIS

1 Start from the result we just derived:

$$a = \frac{v^2}{r}$$

2 Substitute $v = \omega r$:

$$a = \frac{(\omega r)^2}{r} = \frac{\omega^2 r^2}{r} \quad \Rightarrow \quad \boxed{a = r\omega^2}$$

Step 4 of the previous derivation also gave the handy middle form $a = v\omega$.

Centripetal acceleration FORMULA SHEET

$$a = \frac{v^2}{r}$$

a: acceleration towards centre / m s^{-2}

Centripetal acceleration FORMULA SHEET

$$a = r\omega^2$$

ω: angular velocity / rad s^{-1}

COMMON MISTAKE

"Constant speed, so no acceleration" is the classic wrong answer. In uniform circular motion the **speed** is constant but the **velocity** is not, so there is an acceleration, of size v^2/r, always pointing at the centre.

WORKED EXAMPLE

A washing machine drum of radius 0.25 m spins at 1200 revolutions per minute. Calculate the speed of a sock pressed against the drum wall and its centripetal acceleration. How many times larger than g is this?

Show solution

Convert rpm to rad s⁻¹: each revolution is 2π rad, and there are 60 s in a minute.

$$\omega = \frac{1200 \times 2\pi}{60} = 126 \text{ rad s}^{-1}$$

$$v = \omega r = 126 \times 0.25 = 31.4 \text{ m s}^{-1}$$

$$a = r\omega^2 = 0.25 \times 126^2 = 3.9 \times 10^3 \text{ m s}^{-2} \approx 400\,g$$

The drum wall pushes the sock towards the centre with a huge force. Water in the sock is not held by the wall at the holes, so it carries on in a straight line, out through the holes.

Centripetal force

Newton's first law says an object keeps moving in a straight line at constant speed unless a resultant force acts on it. So to keep something going round a circle, there must be a **resultant force** on it, and by Newton's second law it points the same way as the acceleration: **towards the centre**.

Centripetal force

FORMULA SHEET

$$F = ma = \frac{mv^2}{r}$$

F: resultant force towards centre / N

m: mass / kg

Centripetal force

FORMULA SHEET

$$F = mr\omega^2$$

r: radius / m ω: angular velocity / rad s⁻¹

DEFINITION

Centripetal force

The **centripetal force** is the **resultant force** on an object moving in a circle. It is directed towards the centre of the circle, and it is needed both to **produce** circular motion and to **maintain** it.

KEY IDEA

Centripetal force is a job, not a new force

"Centripetal" describes the direction of a resultant force. Something real has to supply it, and you should never draw an extra arrow labelled "centripetal force" on a free-body diagram.

Situation	What provides the centripetal force
Puck on a string on an air table	Tension in the string
Car going round a flat bend	Friction from the road on the tyres
The Moon orbiting the Earth	Gravitational force from the Earth
Clothes in a spin dryer	Normal contact force from the drum wall
Electron in a uniform magnetic field	Magnetic force Bqv
Electron round a nucleus (simple model)	Electric (Coulomb) attraction

Because the centripetal force is always at 90° to the velocity, it never pushes the object forwards or backwards along its path. It does **no work**, so it changes the direction of motion but not the speed or the kinetic energy.

COMMON MISTAKE

There is no outward force

When a car turns left you feel "thrown" to the right. Nothing is pushing you right. Your body is trying to carry on in a straight line (Newton's first law) while the car turns underneath you, until the door pushes you **inwards**, towards the centre of the turn. If the string breaks, a ball flies off along the tangent, not radially outwards.

WORKED EXAMPLE

A 0.20 kg puck on a frictionless table is whirled in a horizontal circle of radius 0.50 m on a string, making 2.0 revolutions per second. (a) Calculate the tension in the string. (b) The string snaps at 25 N. What is the greatest number of revolutions per second it can make?

Show solution

(a) $\omega = 2\pi f = 2\pi \times 2.0 = 12.6 \text{ rad s}^{-1}$. The tension is the only horizontal force, so it is the centripetal force:

$$T = mr\omega^2 = 0.20 \times 0.50 \times 12.6^2 = 15.8 \text{ N}$$

(b) Rearrange for ω with $T = 25$ N:

$$\omega = \sqrt{\frac{T}{mr}} = \sqrt{\frac{25}{0.20 \times 0.50}} = 15.8 \text{ rad s}^{-1}, \qquad f = \frac{\omega}{2\pi} = 2.5 \text{ revolutions per second}$$

Cars on bends and banked tracks

On a flat road, the only horizontal force on a car is **friction** from the road on the tyres, pointing towards the centre of the bend. The friction needed is

$$F = \frac{mv^2}{r}$$

The v^2 is the important part. Double your speed round a bend and you need **four times** as much friction. A tighter bend (smaller r) also needs more. The road can only supply so much grip, and it supplies much less when it's wet or icy. If the friction needed is more than the road can give, the car cannot follow the bend: it moves on a wider curve, closer to a straight line, and slides towards the outside of the bend.

WORKED EXAMPLE

A car of mass 1200 kg goes round a flat bend of radius 50 m at 15 m s^{-1}. (a) What friction force is needed? (b) The greatest friction the road can provide is 7200 N. What is the maximum safe speed?

Show solution

$$F = \frac{mv^2}{r} = \frac{1200 \times 15^2}{50} = 5400 \text{ N}$$

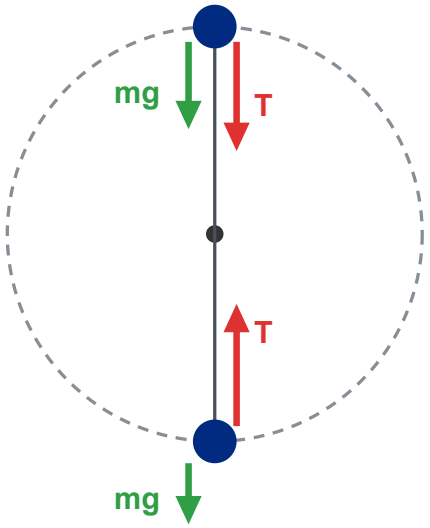
(b) Rearrange with $F = 7200$ N:

$$v = \sqrt{\frac{Fr}{m}} = \sqrt{\frac{7200 \times 50}{1200}} = 17.3 \text{ m s}^{-1}$$

Only 15% faster than before uses up all the grip, because $F \propto v^2$: $1.15^2 \approx 1.33$, and $5400 \times 1.33 = 7200$.

Vertical circles

Now swing a ball on a string in a **vertical** circle, or ride a roller coaster round a loop. Nothing new is needed: at every point, the resultant force towards the centre must equal mv^2/r. What changes is that the weight now sometimes points towards the centre and sometimes away from it, and the speed changes as the ball rises and falls (kinetic energy is traded for gravitational potential energy).



At the top

both forces point to the centre:

T + mg = mv²/r

At the bottom

weight points away from the centre:

T − mg = mv²/r

Free-body diagrams at the top and bottom of a vertical circle. Take "towards the centre" as positive.

- **At the bottom:** $T = mg + \dfrac{mv^2}{r}$. The string has to hold the ball up *and* provide the centripetal force, so the tension is greatest here. This is where a string is most likely to snap, and where you feel heaviest on a ride.
- **At the top:** $T = \dfrac{mv^2}{r} - mg$. Gravity is already helping, so less tension is needed. The ball is also moving slowest here.
- **Minimum speed at the top:** the tension can't be negative (a string can pull but not push). The smallest possible speed is when $T = 0$ and gravity alone provides the centripetal force: $mg = mv^2/r$, so $v_{\min} = \sqrt{gr}$. Any slower, and the string goes slack and the ball falls inside the circle as a projectile.

The same reasoning explains why water stays in a bucket swung over your head: if $v \geq \sqrt{gr}$ at the top, the bucket has to push the water down to keep it on the circle, so the water stays pressed into the bucket.

WORKED EXAMPLE

A roller coaster car (total mass 500 kg) enters a vertical loop of radius 8.0 m at 22 m s⁻¹ at the bottom. Ignore friction. (a) What is the minimum speed needed at the top? (b) Find the speed at the top. (c) Find the normal contact force from the track on the car at the top. (d) What force does the seat exert on a 60 kg rider at the bottom?

Show solution

(a) At the minimum speed the track exerts no force and weight alone provides the centripetal force:

$$v_{\min} = \sqrt{gr} = \sqrt{9.81 \times 8.0} = 8.86 \text{ m s}^{-1}$$

(b) The top is $2r = 16$ m above the bottom. Energy is conserved, so $\frac{1}{2}mv_{\text{top}}^2 = \frac{1}{2}mv_{\text{bottom}}^2 - mg(2r)$. Cancel m:

$$v_{\text{top}}^2 = 22^2 - 2 \times 9.81 \times 16 = 484 - 314 = 170, \qquad v_{\text{top}} = 13.0 \text{ m s}^{-1}$$

(c) At the top, the normal force N and the weight both point down, towards the centre:

$$N + mg = \frac{mv^2}{r} \quad \Rightarrow \quad N = \frac{500 \times 170}{8.0} - 500 \times 9.81 = 10\,600 - 4900 = 5.7 \times 10^3 \text{ N}$$

(d) At the bottom the seat pushes up and the weight acts down:

$$N = mg + \frac{mv^2}{r} = 60 \times 9.81 + \frac{60 \times 22^2}{8.0} = 589 + 3630 = 4.2 \times 10^3 \text{ N}$$

EXAM TIP

For every circular motion question, use the same three steps: (1) draw the real forces, (2) find the **resultant towards the centre**, (3) set it equal to mv^2/r or $mr\omega^2$. If the speed changes (vertical circles), use conservation of energy to find v at the point you need.

Quick checks

QUICK CHECK

What is 1.5 rad in degrees?

- 27°
- 86°
- 270°
- 0.026°

$1.5 \times \frac{180}{\pi} = 85.9°$. Quick check: 1 rad is about 57°, so 1.5 rad is a bit less than 90°. 0.026 is what you get converting the wrong way.

QUICK CHECK

A car drives round a roundabout at a constant speed. Which statement is correct?

- Its velocity is constant.
- Its acceleration is zero.
- Its acceleration is directed towards the centre of the roundabout.
- The resultant force on it is directed away from the centre.

The direction of the velocity keeps changing, so the car accelerates, towards the centre, with $a = v^2/r$. The resultant force points the same way as the acceleration: inwards.

QUICK CHECK

Two children sit on a spinning roundabout, one 1.0 m from the centre and one 2.0 m from the centre. Compared with the inner child, the outer child has

- the same angular velocity and twice the speed.
- twice the angular velocity and the same speed.
- the same speed and the same acceleration.
- half the centripetal acceleration.

They go round together, so ω is the same. $v = \omega r$ doubles, and $a = r\omega^2$ also doubles (not halves).

QUICK CHECK

A car goes round a bend of fixed radius. If its speed doubles, the friction force needed

- doubles.
- becomes four times bigger.
- halves.
- stays the same.

$F = mv^2/r$, so $F \propto v^2$: $2^2 = 4$.

QUICK CHECK

A ball on a string is swung in a vertical circle. Where is the tension in the string greatest?

- At the top
- At the bottom
- Halfway up the sides
- It is the same everywhere

At the bottom the ball is fastest, so it needs the most centripetal force, and the string must also cancel the weight: $T = mg + mv^2/r$.

QUICK CHECK

A puck whirled in a horizontal circle on a frictionless table is moving due north when the string breaks. Which way does it move next?

- Radially outwards, away from the centre
- Due north, in a straight line
- In a curve that gradually straightens
- Towards the centre

Once the string breaks there is no horizontal force, so by Newton's first law it carries on in a straight line with the velocity it had at that instant, which was along the tangent.

A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Electric Fields

Rub a balloon on your jumper and it can lift your hair without touching it: a charged object fills the space around it with an **electric field**. In this topic we define field strength, use Coulomb's law for the force between two charges, meet electric potential, and see how field and potential describe the same thing in two different ways. These ideas run straight on into capacitors, particle accelerators and the deflection of electron beams.

Electric fields and field strength

DEFINITION

Electric field

An **electric field** is a region in which a **charged particle experiences a force**. It is a force field, like a gravitational field, except that it acts on charge instead of mass.

To measure how strong a field is at some point, imagine placing a small positive "test" charge there and measuring the force on it. Twice the charge feels twice the force, so the useful quantity is the force on each coulomb: the **electric field strength**.

Electric field strength

FORMULA SHEET

$$E = \frac{F}{Q}$$

E: electric field strength / N C^{-1} F: force on the charge / N Q: charge / C

DEFINITION

Electric field strength

The **electric field strength** at a point is the **force per unit charge** on a small positive test charge placed at that point. It is a vector, pointing in the direction of the force on a positive charge, and is measured in N C^{-1}.

- A **positive** charge is pushed **along** the field direction.
- A **negative** charge, such as an electron, is pushed the **opposite** way to the field.
- The test charge has to be small so that it doesn't push the charges that make the field around and change the very thing you're measuring.

WORKED EXAMPLE

An electron is in a uniform electric field of strength $5.0 \times 10^4\ \text{N C}^{-1}$. Calculate the force on it and its acceleration, and compare the force with its weight.

Show solution

$$F = EQ = 5.0 \times 10^4 \times 1.60 \times 10^{-19} = 8.0 \times 10^{-15}\ \text{N}$$

$$a = \frac{F}{m} = \frac{8.0 \times 10^{-15}}{9.11 \times 10^{-31}} = 8.8 \times 10^{15}\ \text{m s}^{-2}$$

The weight is $mg = 9.11 \times 10^{-31} \times 9.81 = 8.9 \times 10^{-30}\ \text{N}$, about 10^{15} times smaller than the electric force. That's why gravity is ignored for electrons and ions in electric fields. The force acts in the opposite direction to the field, because the electron is negative.

Coulomb's law

Two point charges push each other apart if they are alike and pull together if they are opposite. Coulomb found that the size of the force depends on the charges and the distance between them in a very simple way.

DEFINITION

Coulomb's law

The force between two point charges is **proportional to the product of the charges** and **inversely proportional to the square of the distance** between them.

Coulomb's law

FORMULA SHEET

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

Q_1, Q_2: the two charges / C r: distance between their centres / m

ε_0: permittivity of free space, $8.85 \times 10^{-12}\ \text{F m}^{-1}$

The constant at the front is often written $k = \dfrac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9\ \text{N m}^2\ \text{C}^{-2}$. It is enormous, which tells you that a coulomb is a huge amount of charge: two charges of 1 C held 1 m apart would push on each other with about 9

$\times 10^9$ N. Everyday static charges are microcoulombs or nanocoulombs.

- **Sign:** if Q_1Q_2 is positive (like charges), F is positive and the force is **repulsive**. If it is negative (unlike charges), the force is **attractive**.
- **Newton's third law:** each charge feels the same size of force, in opposite directions, even if one charge is much bigger than the other.
- **Inverse square:** double the distance and the force becomes $\frac{1}{4}$; triple it and the force becomes $\frac{1}{9}$.

EXAM TIP

Three classic slips: forgetting to **square** r; using the distance between the surfaces of two spheres instead of their **centres**; and not converting μC (10^{-6} C) or nC (10^{-9} C). Do the sign separately: work out the size of the force, then decide "attract" or "repel" from the charges.

WORKED EXAMPLE

In a simple model of a hydrogen atom, the electron orbits the proton at a distance of 5.3×10^{-11} m. (a) Calculate the electric force between them. (b) Compare it with the gravitational force, 3.6×10^{-47} N. (c) The electric force provides the centripetal force. Find the electron's speed.

Show solution

(a) Both charges have size $e = 1.60 \times 10^{-19}$ C:

$$F = \frac{8.99 \times 10^9 \times (1.60 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} = \frac{2.30 \times 10^{-28}}{2.81 \times 10^{-21}} = 8.2 \times 10^{-8} \text{ N}$$

The charges are opposite, so the force is attractive.

(b) $8.2 \times 10^{-8} / 3.6 \times 10^{-47} \approx 2 \times 10^{39}$. Gravity is completely negligible inside atoms.

(c) Using circular motion, $F = mv^2/r$:

$$v = \sqrt{\frac{Fr}{m}} = \sqrt{\frac{8.2 \times 10^{-8} \times 5.3 \times 10^{-11}}{9.11 \times 10^{-31}}} = 2.2 \times 10^6 \text{ m s}^{-1}$$

That is under 1% of the speed of light, so the non-relativistic formulas are fine here.

The field of a point charge

Combine Coulomb's law with the definition of field strength and you get the field at any distance from a single charge.

E around a point charge

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Place a small positive test charge q a distance r from a point charge Q. Coulomb's law gives the force on it:

$$F = \frac{Qq}{4\pi\varepsilon_0 r^2}$$

2. Field strength is force per unit charge, so divide by q:

$$E = \frac{F}{q} = \frac{Qq}{4\pi\varepsilon_0 r^2\, q} \quad \Rightarrow \quad \boxed{E = \frac{Q}{4\pi\varepsilon_0 r^2}}$$

The test charge cancels: the field belongs to Q alone, whether or not anything is there to feel it.

Field strength of a point charge

FORMULA SHEET

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

Q: charge making the field / C r: distance from the charge / m

This is a **radial** field: the field lines point straight out from a positive charge (and straight in towards a negative one), and they spread apart as r increases, showing that the field weakens with the square of the distance. Outside a charged sphere, the field is the same as if all the charge were concentrated at its centre, so you measure r from the centre.

WORKED EXAMPLE

The dome of a Van de Graaff generator is a sphere of radius 0.15 m carrying a charge of $+2.0\ \mu\text{C}$. Calculate the field strength at its surface and 0.30 m from its centre.

Show solution

$$E_{0.15} = \frac{8.99 \times 10^{9} \times 2.0 \times 10^{-6}}{0.15^{2}} = 8.0 \times 10^{5}\ \text{N C}^{-1}$$

$$E_{0.30} = \frac{8.99 \times 10^{9} \times 2.0 \times 10^{-6}}{0.30^{2}} = 2.0 \times 10^{5}\ \text{N C}^{-1}$$

Doubling the distance from the centre quarters the field. Notice that 0.30 m from the centre is only 0.15 m from the surface: always measure from the centre.

Electric potential

Field strength tells you the force on each coulomb. **Potential** tells you the energy of each coulomb. Imagine pushing a small positive charge in from very far away towards a positive charge Q. It is repelled, so you must do work on it, and that work is stored as electric potential energy. The work done per coulomb is the potential at the point where you stop.

DEFINITION

Electric potential

The **electric potential** at a point is the **work done per unit charge** in bringing a small positive test charge from **infinity** to that point. It is a **scalar**, measured in volts (1 V = 1 J C^{-1}), and is zero at infinity.

Potential in a radial field

FORMULA SHEET

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

V: electric potential / V Q: charge / C r: distance from the charge / m

- Near a **positive** charge the potential is **positive** (you had to push your test charge in) and falls towards zero as r increases.
- Near a **negative** charge the potential is **negative** (the test charge is pulled in, so energy is given out) and rises towards zero as r increases.
- Notice the powers: $V \propto 1/r$ but $E \propto 1/r^{2}$. Double the distance and the potential halves but the field quarters.

- The **potential difference** between two points is the work done per unit charge moving between them, so the work done moving charge Q through a p.d. ΔV is $W = Q\,\Delta V$, exactly as in Unit 2.

An **equipotential** is a line (or surface) joining points at the same potential. Moving a charge along an equipotential takes no work, so the field can have no component along it: **equipotentials are always at right angles to field lines**. Around a point charge they are concentric spheres. Drawn at equal steps of potential, they get further apart as you move out, because the field is getting weaker.

How field and potential are related

Field and potential are two descriptions of the same field: one tells you about forces, the other about energies. The link between them is the gradient of the potential.

Field strength is minus the potential gradient

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Move a small positive charge q a short distance Δr in the direction of the field E. The field pushes it with force qE, so the field does work on it:

$$W = qE\,\Delta r$$

2. That energy comes out of the charge's electric potential energy, which falls by $qE\,\Delta r$. Per unit charge, the potential falls by $E\,\Delta r$:

$$\Delta V = -E\,\Delta r$$

3. Rearrange:

$$\boxed{E = -\frac{\Delta V}{\Delta r}}$$

The field strength is the gradient of the V–r graph with its sign reversed. The minus sign means the field points "downhill", from high potential to low potential.

KEY IDEA

Reading the two graphs

- The **gradient** of a V–r graph at any point is $-E$. Where the curve is steepest, the field is strongest.
- Equipotentials close together mean a steep potential gradient, which means a strong field.
- Field lines always point from higher potential to lower potential.

- Units check: the gradient is in V m⁻¹, and $1\ \text{V m}^{-1} = 1\ \text{J C}^{-1}\,\text{m}^{-1} = 1\ \text{N m C}^{-1}\,\text{m}^{-1} = 1\ \text{N C}^{-1}$.

WORKED EXAMPLE

(a) Calculate the potential at the surface of the Van de Graaff dome from the previous example (radius 0.15 m, charge $+2.0\ \mu\text{C}$) and 0.30 m from its centre. (b) A proton moves from a point where the potential is 60 kV to a point where it is 20 kV. Starting from rest, how fast is it moving?

Show solution

(a)

$$V_{0.15} = \frac{8.99 \times 10^{9} \times 2.0 \times 10^{-6}}{0.15} = 1.2 \times 10^{5}\ \text{V}, \qquad V_{0.30} = 6.0 \times 10^{4}\ \text{V}$$

Doubling the distance halved the potential but quartered the field. For a point charge, $E = V/r$: $1.2 \times 10^{5}/0.15 = 8.0 \times 10^{5}\ \text{N C}^{-1}$, matching the earlier answer.

(b) The proton falls through a potential difference of $60 - 20 = 40\ \text{kV}$. The work done on it by the field becomes kinetic energy:

$$E_k = Q\,\Delta V = 1.60 \times 10^{-19} \times 4.0 \times 10^{4} = 6.4 \times 10^{-15}\ \text{J}$$

$$v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 6.4 \times 10^{-15}}{1.67 \times 10^{-27}}} = 2.8 \times 10^{6}\ \text{m s}^{-1}$$

Uniform fields between parallel plates

Connect two parallel metal plates to a supply and one becomes positive and the other negative. Between them (away from the edges) the field lines are **parallel, equally spaced and at right angles to the plates**: the field is **uniform**, the same strength and direction everywhere. Near the edges the lines bulge outwards; these are called **edge effects**.

E = V/d for parallel plates

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Move a positive charge Q from the negative plate to the positive plate, a distance d, against the field. The force needed is $F = EQ$, which is constant because the field is uniform. The work done is

$$W = Fd = EQd$$

2. By the definition of potential difference, moving charge Q through p.d. V takes work

$$W = QV$$

3. The two must be equal. Q cancels:

$$EQd = QV \quad \Rightarrow \quad \boxed{E = \frac{V}{d}}$$

So the unit of field strength can also be written V m⁻¹. This is just $E = -\Delta V/\Delta r$ for a field whose potential falls steadily from one plate to the other.

Uniform field between parallel plates

FORMULA SHEET

$$E = \frac{V}{d}$$

V: p.d. between the plates / V d: plate separation / m

The potential falls at a steady rate from the positive plate to the negative plate, so the **equipotentials are planes parallel to the plates and equally spaced**. Halfway across, the potential is halfway between the two plate potentials. Notice that E depends only on V and d, not on the area of the plates or where between them you are.

Charged particles crossing a uniform field

Fire an electron horizontally between horizontal plates. The force eE is constant and vertical, so this is exactly like a projectile under gravity: the **horizontal velocity stays constant** and there is a **constant vertical acceleration** $a = eE/m$. The path is a **parabola**. Once the electron leaves the plates, the force stops and it carries on in a straight line.

With horizontal speed v_0 and plate length L, the time between the plates is $t = L/v_0$, and the sideways deflection at the exit is $y = \frac{1}{2}at^2 = \dfrac{eVL^2}{2md\,v_0^2}$. Bigger V bends it more; a faster electron spends less time in the field and bends less.

WORKED EXAMPLE

An electron travelling at 3.0×10^7 m s^{-1} enters the space midway between two parallel plates 8.0 cm long and 2.0 cm apart, moving parallel to them. The p.d. between the plates is 150 V. Calculate (a) the field strength, (b) the acceleration of the electron, (c) its deflection as it leaves the plates and (d) the angle its path then makes with its original direction.

Show solution

(a)

$$E = \frac{V}{d} = \frac{150}{0.020} = 7500 \text{ V m}^{-1}$$

(b)

$$a = \frac{eE}{m} = \frac{1.60 \times 10^{-19} \times 7500}{9.11 \times 10^{-31}} = 1.32 \times 10^{15} \text{ m s}^{-2}$$

(c) Horizontally the speed is constant: $t = L/v_0 = 0.080/3.0 \times 10^7 = 2.67 \times 10^{-9}$ s. Vertically it starts from rest:

$$y = \tfrac{1}{2}at^2 = \tfrac{1}{2} \times 1.32 \times 10^{15} \times (2.67 \times 10^{-9})^2 = 4.7 \times 10^{-3} \text{ m} = 4.7 \text{ mm}$$

That is less than the 10 mm gap to the plate, so the electron gets through.

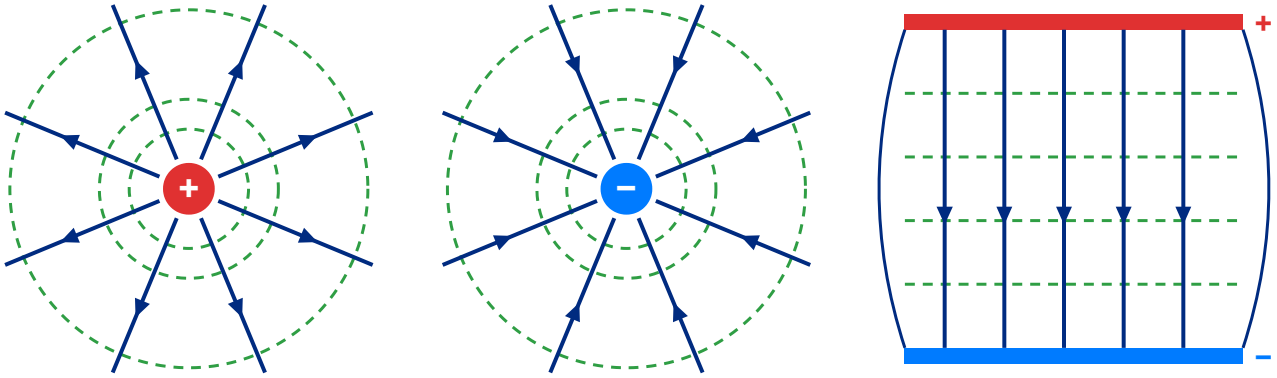
(d) Vertical velocity at the exit: $v_y = at = 1.32 \times 10^{15} \times 2.67 \times 10^{-9} = 3.5 \times 10^6$ m s^{-1}.

$$\theta = \tan^{-1}\left(\frac{3.5 \times 10^6}{3.0 \times 10^7}\right) = 6.7^\circ$$

Drawing field lines and equipotentials

Exam questions often ask you to sketch or interpret field patterns. The rules are few:

- Field lines start on **positive** charges and end on **negative** charges (or go off to infinity). The arrow shows the direction of the force on a positive charge.
- Field lines **never cross**: the field can only point one way at any point.
- The **closer** the lines, the **stronger** the field. Parallel, equally spaced lines mean a uniform field.
- Field lines meet the surface of a conductor at **90°**.
- Equipotentials are always **perpendicular** to field lines. Drawn at equal steps of potential, they are closer together where the field is stronger.



positive point charge | negative point charge | uniform field between plates

Field lines (solid, with arrows) and equipotentials (green, dashed). The radial equipotentials are drawn at equal steps of potential, so they spread out as the field weakens. Between the plates they are equally spaced because the field is uniform.

COMMON MISTAKE

A field line is **not** the path a charge will follow. It shows the direction of the force, and so of the **acceleration**. A charge that is already moving, or one released in a curved field, will not stay on a field line, for the same reason a thrown ball doesn't move in the direction of gravity.

KEY IDEA

Electric and gravitational fields side by side

	Electric	Gravitational
Acts on	charge	mass
Field strength	$E = F/Q$, N C^{-1}	$g = F/m$, N kg^{-1}
Force between points	$Q_1Q_2/4\pi\varepsilon_0 r^2$: attract or repel	Gm_1m_2/r^2: always attract
Potential	$Q/4\pi\varepsilon_0 r$: positive or negative	$-Gm/r$: always negative
Uniform field	between parallel plates	close to the Earth's surface

Both are inverse-square fields, which is why the maths is identical. You'll meet the gravitational versions on the Gravitational Fields page in Unit 5.

Quick checks

QUICK CHECK

The distance between two point charges is doubled. What happens to the force between them?

- It halves.
- It falls to a quarter.
- It doubles.
- It falls to an eighth.

$F \propto 1/r^2$, so doubling r gives $1/2^2 = 1/4$ of the force. "Halves" would be the answer for the potential, which goes as $1/r$.

QUICK CHECK

Which of these is a unit of electric field strength?

- V m^{-1}
- V m
- N C
- J C^{-1}

From $E = V/d$, field strength is in V m^{-1}, which is the same as N C^{-1}. J C^{-1} is the volt, the unit of potential.

QUICK CHECK

The electric field at a point points due east. Which way is the force on an electron placed there?

- East
- West
- There is no force on an electron at rest.
- At right angles to the field

The field direction is the direction of the force on a **positive** charge. The electron is negative, so it is pushed the opposite way. Electric forces act on charges whether or not they are moving.

QUICK CHECK

Equipotentials around a point charge are drawn at equal steps of potential. Moving away from the charge, they

- are equally spaced.
- get further apart.
- get closer together.
- are straight, parallel lines.

The field gets weaker further out, so the potential changes more slowly with distance: you have to go further to drop by the same number of volts.

QUICK CHECK

The p.d. across a pair of parallel plates is kept the same while their separation is doubled. What happens to the force on a charge between them?

- It doubles.
- It halves.
- It stays the same.
- It falls to a quarter.

$E = V/d$, so doubling d halves E, and $F = EQ$ halves too. It is not an inverse-square effect: the field between plates is uniform.

QUICK CHECK

Point P is midway between a charge $+Q$ and a charge $-Q$. Which statement is correct at P?

- $V = 0$ and $E = 0$
- $V = 0$ but E is not zero; it points towards $-Q$.
- $E = 0$ but V is not zero.
- V and E both point towards $+Q$.

Potential is a scalar: $+kQ/r$ and $-kQ/r$ add to zero. Field is a vector: both charges' fields at P point the same way (away from $+Q$, towards $-Q$), so they add rather than cancel. Potential has no direction at all.

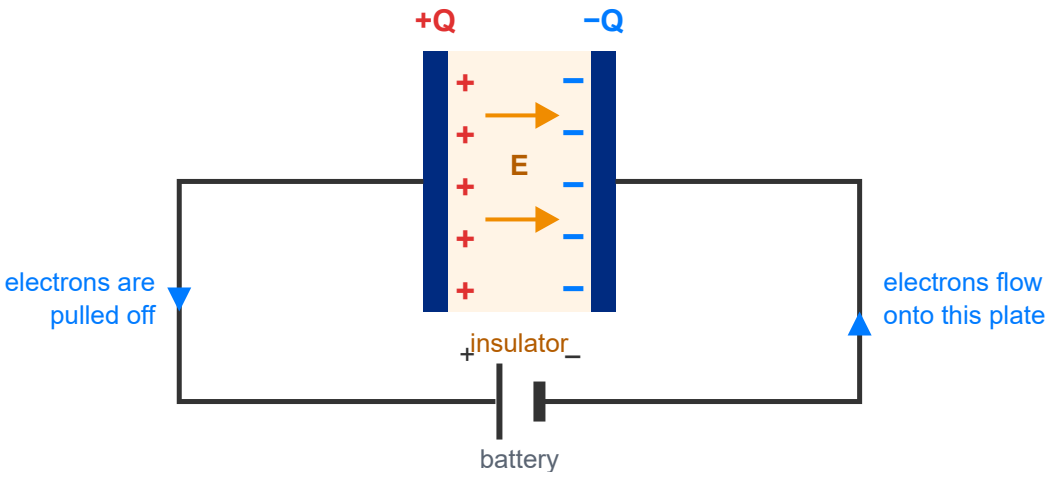
A2 · Unit 4 · WPH14 | Further Mechanics, Fields & Particles

Capacitors

A capacitor stores energy by holding positive and negative charge apart. It stores far less energy than a battery, but it can hand all of it back in a fraction of a second, which is why camera flashes, defibrillators and the smoothing circuit in every phone charger use one. On this page you will see where the ½ in $W = \frac{1}{2}QV$ comes from, why capacitors charge and discharge exponentially, and how a log graph turns that curve into a straight line.

What a capacitor is

The simplest capacitor is two parallel metal plates with an **insulator** between them (air, paper, plastic or a thin oxide layer). Connect it to a battery and the battery pushes electrons onto one plate and pulls the same number of electrons off the other. No charge ever crosses the gap. One plate ends up with charge $+Q$, the other with $-Q$.



Charging: the battery moves electrons round the circuit from the left plate to the right plate. The plates end up with equal and opposite charges, and an electric field across the gap.

As charge builds up, the plates develop a p.d. that **opposes** the battery. New electrons arriving at the negative plate are repelled by the ones already there. Charging stops when the p.d. across the capacitor equals the e.m.f. of the battery. At that point there is no p.d. left to drive a current, so the current is zero even though the circuit is complete.

COMMON MISTAKE

A capacitor does not store **net** charge: $+Q$ on one plate and $-Q$ on the other add up to zero. What it stores is **energy**, by keeping the charges separated. When a question says "the charge stored is Q" it means the charge on **one** plate.

Capacitance: C = Q/V

Push twice as much charge onto the plates and the field between them doubles, so the p.d. doubles too. Charge is **proportional** to p.d., and the constant of proportionality tells you how much charge the capacitor holds for each volt. That is its **capacitance**.

Capacitance

FORMULA SHEET

$$C = \frac{Q}{V}$$

C: capacitance / F Q: charge on one plate / C V: p.d. across the capacitor / V

DEFINITION

Capacitance is the charge stored per unit potential difference across the capacitor, $C = Q/V$. The unit is the **farad**: a capacitor has a capacitance of 1 F if it stores 1 C of charge when the p.d. across it is 1 V ($1\ \text{F} = 1\ \text{C V}^{-1}$).

One farad is huge. Real capacitors are usually measured in microfarads ($1\ \mu\text{F} = 10^{-6}\ \text{F}$), nanofarads ($10^{-9}\ \text{F}$) or picofarads ($10^{-12}\ \text{F}$). A 470 μF capacitor charged to 9.0 V holds $Q = CV = 470 \times 10^{-6} \times 9.0 = 4.2 \times 10^{-3}\ \text{C}$, a few millicoulombs.

Because $Q \propto V$, a graph of p.d. against charge is a straight line through the origin with gradient $V/Q = 1/C$. A bigger capacitor gives a **shallower** line: it takes more charge to raise its p.d. by each volt.

Energy stored in a capacitor

Charging a capacitor takes work. The first bit of charge goes on easily, because the plates are uncharged and the p.d. is zero. Every later bit has to be pushed onto a plate that already repels it, against a p.d. that keeps growing. The last bit is pushed against the full p.d. V.

Moving a small charge ΔQ through a p.d. V takes work $\Delta W = V\,\Delta Q$. On a graph of V against Q, that is the area of a thin strip of height V and width ΔQ. Add up all the strips and you get the **area under the graph**, which is a triangle.

Deriving W = ½QV from the area under the V–Q graph

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. To move a small extra charge ΔQ onto the plates when the p.d. is V, you do work (p.d. is work done per unit charge):

$$\Delta W = V\,\Delta Q$$

If ΔQ is small, V hardly changes while you add it.

2. On a graph of V (vertical) against Q (horizontal), $V\,\Delta Q$ is the area of a thin strip under the line. The total work done charging from 0 to Q is the sum of all the strips, which is the area under the graph:

$$W = \sum V\,\Delta Q = \text{area under the } V\text{–}Q \text{ graph}$$

3. Because $V \propto Q$, the graph is a straight line through the origin, so the area is a triangle with base Q and height V:

$$W = \tfrac{1}{2} \times \text{base} \times \text{height} \quad \Rightarrow \quad \boxed{W = \tfrac{1}{2}QV}$$

The ½ is there because, on average, each coulomb was pushed on against only half of the final p.d.

Deriving W = ½CV² and W = ½Q²/C

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the result above and the definition of capacitance:

$$W = \tfrac{1}{2}QV, \qquad C = \frac{Q}{V}$$

2. To get rid of Q, write $Q = CV$ and substitute:

$$W = \tfrac{1}{2}(CV)V \quad \Rightarrow \quad \boxed{W = \tfrac{1}{2}CV^2}$$

3. To get rid of V instead, write $V = Q/C$ and substitute:

$$W = \tfrac{1}{2}Q\left(\frac{Q}{C}\right) \quad \Rightarrow \quad \boxed{W = \frac{1}{2}\frac{Q^2}{C}}$$

All three are the same energy. Pick the one that uses the two quantities you already know.

Energy stored

FORMULA SHEET

$$W = \tfrac{1}{2}QV$$

Energy stored

FORMULA SHEET

$$W = \tfrac{1}{2}CV^2$$

Energy stored

FORMULA SHEET

$$W = \frac{1}{2}\frac{Q^2}{C}$$

EXAM TIP

For a given capacitor the energy goes as V^2: double the p.d. and you store **four** times the energy. This is why defibrillator and camera-flash capacitors are charged to hundreds of volts.

COMMON MISTAKE

$W = QV$ is the work done by a battery moving charge Q through a constant p.d. V. For a capacitor the p.d. rises from zero as the charge goes on, so the energy stored is only $\tfrac{1}{2}QV$. Don't drop the half.

WORKED EXAMPLE

A 220 μF capacitor is charged to a p.d. of 9.0 V. Calculate (a) the charge stored and (b) the energy stored. (c) The p.d. is increased to 18 V. How much extra energy is stored?

Show solution

(a) $Q = CV = 220 \times 10^{-6} \times 9.0 = 1.98 \times 10^{-3}\ \text{C} \approx 2.0\ \text{mC}$

(b) $W = \tfrac{1}{2}CV^2 = \tfrac{1}{2} \times 220 \times 10^{-6} \times 9.0^2 = 8.9 \times 10^{-3}\ \text{J}$

(c) Energy goes as V^2, so doubling V gives four times the energy: $4 \times 8.91 \times 10^{-3} = 3.56 \times 10^{-2}\ \text{J}$. The extra energy is $3.56 \times 10^{-2} - 0.89 \times 10^{-2} = 2.7 \times 10^{-2}\ \text{J}$ (three times the original, not one).

Charging and discharging through a resistor

In real circuits a capacitor charges and discharges through a resistor, and the resistor sets how fast it happens. The key to every graph in this topic is to ask: **what is the p.d. across the resistor right now?** That p.d. decides the current, $I = V_R/R$.

Charging

At the instant the switch closes, the capacitor is empty, so $V_C = 0$ and the whole battery e.m.f. is across the resistor. The current starts at its biggest value, $I_0 = V_0/R$. As charge builds up, V_C rises. Kirchhoff's second law says $V_0 = V_C + V_R$, so V_R has to fall, and so does the current. Less current means charge arrives more slowly, so V_C rises more slowly. The capacitor creeps towards V_0 but the closer it gets, the slower it goes.

Discharging

Now switch the battery out and connect the charged capacitor across the resistor alone. The capacitor drives a current round the circuit, in the **opposite** direction to the charging current. The p.d. across the resistor is now the p.d. across the capacitor, so $I = V_C/R = Q/RC$. The current is proportional to the charge that is left. A full capacitor empties quickly; a nearly empty one has hardly any p.d. to push with and empties slowly.

"The rate of loss is proportional to the amount left" is the signature of **exponential decay**. You will meet the same pattern in radioactive decay in Unit 5.

KEY IDEA

The shapes to remember

	Charging	Discharging
Charge Q and p.d. V_C	rise quickly, then level off at $Q_0 = CV_0$ and V_0	fall exponentially towards zero
Current I	starts at V_0/R and falls exponentially to zero	starts at V_0/R in the opposite direction and falls exponentially to zero
p.d. across R	$V_0 - V_C$, falls to zero	equal to V_C

Q and V_C always have the same shape because $V_C = Q/C$. The current is the gradient of the Q–t graph ($I = \Delta Q/\Delta t$), so it is largest where the Q–t graph is steepest.

COMMON MISTAKE

When you sketch the discharge current, check whether the question wants its **magnitude** (drawn above the axis) or its **direction** too (drawn below the axis, because it flows the opposite way to the charging current). Also, changing R does not change the final charge $Q_0 = CV_0$. It only changes how long it takes to get there.

The time constant RC

How fast all of this happens depends on just one number, the product RC. A bigger resistance means a smaller current, so charge moves more slowly. A bigger capacitance means more charge has to move for every volt of change. Either way, it takes longer. RC is called the **time constant**, and it really is a time:

$$[RC] = \Omega \times \mathrm{F} = \frac{\mathrm{V}}{\mathrm{A}} \times \frac{\mathrm{C}}{\mathrm{V}} = \frac{\mathrm{C}}{\mathrm{A}} = \frac{\mathrm{A\ s}}{\mathrm{A}} = \mathrm{s}$$

DEFINITION

The **time constant** RC of a capacitor–resistor circuit is the time taken for the charge (or current, or p.d.) of a discharging capacitor to fall to $1/e$ of its initial value, which is about **37%**. While charging from empty, it is the time for the charge to reach $1 - 1/e$, about **63%**, of its final value.

Time	Fraction left when discharging, $e^{-t/RC}$	Fraction reached when charging
RC	0.368 (37%)	0.632 (63%)
$2RC$	0.135	0.865
$3RC$	0.050	0.950
$5RC$	0.007	0.993

After about $5RC$ a capacitor is, for practical purposes, fully charged or fully discharged. An exponential never actually reaches zero, but after $5RC$ less than 1% is left.

Half-life

An exponential decay loses the **same fraction** in every equal interval of time. So there is a fixed time, the half-life $t_{1/2}$, in which the charge always halves: from 100% to 50%, from 50% to 25%, and so on.

Half-life of a discharging capacitor

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 After one half-life, the charge is half its starting value:

$$\tfrac{1}{2}Q_0 = Q_0\, e^{-t_{1/2}/RC} \quad \Rightarrow \quad \tfrac{1}{2} = e^{-t_{1/2}/RC}$$

2 Take natural logs of both sides. $\ln \tfrac{1}{2} = -\ln 2$:

$$-\ln 2 = -\frac{t_{1/2}}{RC} \quad \Rightarrow \quad \boxed{t_{1/2} = RC \ln 2 \approx 0.69\, RC}$$

The starting charge cancelled out, which is why the half-life is the same however full the capacitor is.

EXAM TIP

Two quick ways to read RC from a discharge graph: find where the curve has fallen to 37% of its starting value, or draw the tangent at $t = 0$ and see where it cuts the time axis. Both give $t = RC$. Using the 37% point is usually more accurate, because drawing a tangent by eye is hard.

The exponential discharge equations

For a capacitor discharging through a resistor, the charge left after time t is given on the formula sheet:

Capacitor discharge

FORMULA SHEET

$$Q = Q_0\, e^{-t/RC}$$

Q_0: charge at $t = 0$ / C t: time since discharge began / s RC: time constant / s

The power $-t/RC$ is "how many time constants have passed", with a minus sign because the charge is falling. Each time constant multiplies what's left by $1/e = 0.368$. The p.d. and current behave in exactly the same way, and you need to be able to show why.

Deriving V = V₀e^(−t/RC) and I = I₀e^(−t/RC)

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the discharge equation and divide both sides by C. Since $V = Q/C$ at every instant:

$$\frac{Q}{C} = \frac{Q_0}{C}\, e^{-t/RC} \quad \Rightarrow \quad \boxed{V = V_0\, e^{-t/RC}}$$

$V_0 = Q_0/C$ is the p.d. at the start. C is a constant, so dividing by it doesn't change the shape.

2. During discharge the capacitor is connected directly across the resistor, so the p.d. across the resistor equals the p.d. across the capacitor. Ohm's law gives the current:

$$I = \frac{V}{R} = \frac{V_0}{R}\, e^{-t/RC}$$

3. V_0/R is the current at the start, I_0, so:

$$\boxed{I = I_0\, e^{-t/RC}}$$

FORMULA SHEET

$$I = I_0\, e^{-t/RC}$$

FORMULA SHEET

$$V = V_0\, e^{-t/RC}$$

Turning the curve into a straight line

Exponential curves are hard to analyse by eye. Taking the natural log of both sides turns them into straight lines, which is the trick behind Core Practical 11.

Deriving the log equations

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Take natural logs of both sides of $Q = Q_0 e^{-t/RC}$:

$$\ln Q = \ln\left(Q_0\, e^{-t/RC}\right)$$

2. The log of a product is the sum of the logs, $\ln(ab) = \ln a + \ln b$:

$$\ln Q = \ln Q_0 + \ln\left(e^{-t/RC}\right)$$

3 $\ln$ undoes e, so $\ln(e^x) = x$:

$$\boxed{\ln Q = \ln Q_0 - \frac{t}{RC}}$$

4 Exactly the same steps on the p.d. and current equations give:

$$\boxed{\ln V = \ln V_0 - \frac{t}{RC}} \qquad \boxed{\ln I = \ln I_0 - \frac{t}{RC}}$$

Compare with $y = c + mx$: plotting $\ln V$ against t gives a straight line with gradient $-1/RC$ and intercept $\ln V_0$.

FORMULA SHEET

$$\ln Q = \ln Q_0 - \frac{t}{RC}$$

FORMULA SHEET

$$\ln I = \ln I_0 - \frac{t}{RC}$$

FORMULA SHEET

$$\ln V = \ln V_0 - \frac{t}{RC}$$

EXAM TIP

To find a **time**, rearrange the log equation: $t = RC\ln(V_0/V)$. Keep V_0/V bigger than 1 and the answer comes out positive. To find a **value** after a time, use the exponential form directly, and work out t/RC first so you don't mistype the power into your calculator.

WORKED EXAMPLE

A 470 μF capacitor is charged to 12.0 V and then discharged through a 2.2 kΩ resistor. Calculate (a) the time constant, (b) the initial current, (c) the p.d. after 2.0 s, (d) the time for the p.d. to fall to 3.0 V and (e) the percentage of the original energy still stored at that moment.

Show solution

(a) $RC = 2200 \times 470 \times 10^{-6} = 1.034 \text{ s} \approx 1.03 \text{ s}$

(b) $I_0 = V_0/R = 12.0/2200 = 5.45 \times 10^{-3} \text{ A} = 5.45 \text{ mA}$

(c) $t/RC = 2.0/1.034 = 1.934$, so $V = 12.0\, e^{-1.934} = 12.0 \times 0.1445 = 1.73 \text{ V}$

(d) $t = RC \ln(V_0/V) = 1.034 \times \ln(12.0/3.0) = 1.034 \times 1.386 = 1.43 \text{ s}$

(e) $W \propto V^2$, so the fraction left is $(3.0/12.0)^2 = 1/16 = 6.25\%$. Three quarters of the p.d. has gone, but nearly 94% of the energy has gone.

WORKED EXAMPLE

A camera flash uses a 150 μF capacitor charged to 300 V. When the flash fires, 90% of the stored energy is released in 1.0 ms. Calculate the mean power delivered to the flash tube, and explain why the camera cannot simply use its 3 V battery directly.

Show solution

Energy stored: $W = \frac{1}{2}CV^2 = \frac{1}{2} \times 150 \times 10^{-6} \times 300^2 = 6.75 \text{ J}$

Energy released: $0.90 \times 6.75 = 6.08 \text{ J}$, so mean power $= 6.08/1.0 \times 10^{-3} = 6.1 \times 10^3 \text{ W}$

The battery has enough energy, but its internal resistance limits the current it can deliver, so it can only supply energy slowly (a few watts). The capacitor is charged slowly from the battery over a few seconds, then delivers the energy very quickly through the low resistance of the flash tube (small RC).

Core Practical 11: capacitor discharge

CORE PRACTICAL

Use an oscilloscope or data logger to display and analyse the p.d. across a capacitor as it charges and discharges through a resistor

Method. Connect a capacitor (for example 470 μF), a resistor (for example 47 kΩ), a battery and a two-way switch so that the switch either connects the capacitor to the battery or connects it across the resistor. Connect a voltage sensor and data logger (or an oscilloscope) across the capacitor. Charge the capacitor fully, then flip the switch to discharge it through the resistor, logging the p.d. at regular intervals. Repeat with the switch flipped back to record the charging curve.

Using an oscilloscope. For a small time constant (for example 0.1 μF and 10 kΩ, RC = 1 ms), replace the battery and switch with a square-wave signal generator. The capacitor charges and discharges every cycle and the oscilloscope shows a steady trace. Read V at a series of times from the time-base.

Analysis. Since $\ln V = \ln V_0 - t/RC$, plot $\ln V$ **against** t. A straight line confirms the decay is exponential. The **gradient is** $-1/RC$, so $RC = -1/\text{gradient}$, and with R known, $C = RC/R$. The intercept is $\ln V_0$. You can also check by reading the time for V to fall to 37% of V_0.

- A data logger removes reaction-time error and can take hundreds of readings. If you use a voltmeter and stopwatch instead, choose a large RC (tens of seconds) and take readings at fixed intervals.
- Stop using readings once V is very small: the resolution of the meter becomes a large fraction of the reading and the $\ln V$ points scatter.
- The voltmeter or logger has a finite resistance in parallel with R. If it is not much bigger than R, the capacitor also discharges through the meter, the measured RC is too small and so is C. This is a systematic error: use a high-resistance logger, or a smaller R.
- Electrolytic capacitors have a large tolerance (often ±10% to ±20%), so the measured C can differ from the labelled value by this much.
- **Safety:** connect electrolytic capacitors the right way round and never above their rated voltage, as they can overheat and burst. Discharge large capacitors before handling them.

WORKED EXAMPLE

In Core Practical 11 a student discharges a capacitor through a 47 kΩ resistor and plots $\ln(V/\text{V})$ against t. The line passes through (0 s, 1.79) and (60 s, 0.29). Determine the capacitance, and the p.d. the capacitor was charged to.

Show solution

Gradient $= \dfrac{0.29 - 1.79}{60 - 0} = -0.0250\ \text{s}^{-1}$

Gradient $= -1/RC$, so $RC = 1/0.0250 = 40\ \text{s}$, and $C = \dfrac{40}{47 \times 10^3} = 8.5 \times 10^{-4}\ \text{F} = 850\ \mu\text{F}$

Intercept $= \ln V_0 = 1.79$, so $V_0 = e^{1.79} = 6.0\ \text{V}$

Quick checks

QUICK CHECK

A charged capacitor discharges through a resistor. What fraction of its original charge is left after a time 2RC?

- 0.25
- 0.37
- 0.14
- 0.63

$Q/Q_0 = e^{-2} = 0.135$. Each time constant multiplies the charge by 0.37, so two of them give $0.37^2 \approx 0.14$. 0.25 would be two **half-lives**, and a half-life is shorter than RC.

QUICK CHECK

The p.d. across a capacitor is doubled. What happens to the energy stored?

- It stays the same
- It doubles
- It increases by a factor of 4
- It halves

$W = \frac{1}{2}CV^2$ and C is fixed, so $W \propto V^2$. Doubling V also doubles Q, and $W = \frac{1}{2}QV$ gives the same factor of 4.

QUICK CHECK

Which of these is equivalent to the unit of the time constant, Ω × F?

- A s^{-1}
- s
- C V
- J s^{-1}

$\Omega = \mathrm{V\ A^{-1}}$ and $\mathrm{F = C\ V^{-1}}$, so $\Omega\,\mathrm{F = C\ A^{-1} = (A\ s)\ A^{-1} = s}$.

QUICK CHECK

An uncharged capacitor is connected in series with a resistor and a battery. At the instant the switch is closed, which statement is correct?

- The current is zero because the capacitor has no charge.
- The current is at its maximum, V_0/R.
- The p.d. across the capacitor is V_0.
- The p.d. across the resistor is zero.

With no charge on it, the capacitor has no p.d. across it, so the whole e.m.f. is across the resistor. The current is V_0/R, its largest value. It falls as V_C builds up.

QUICK CHECK

A student plots ln I against t for a capacitor discharging through a resistor. What is the gradient of the line?

- −RC
- ln I_0
- −1/RC
- 1/RC

$\ln I = \ln I_0 - t/RC$ has the form $y = c + mx$ with $m = -1/RC$. The intercept is $\ln I_0$.

QUICK CHECK

A capacitor is charged to a fixed p.d. and discharged through a resistor. The resistor is replaced with one of half the resistance and the experiment is repeated. Which statement is correct?

- The initial current halves and the total charge that flows doubles.
- The time constant doubles.
- The initial current doubles and the total charge that flows is unchanged.
- The energy dissipated in the resistor halves.

$I_0 = V_0/R$ doubles and RC halves, so the discharge is twice as fast. But the capacitor started with the same $Q_0 = CV_0$ and energy $\frac{1}{2}CV_0^2$, and all of it still flows through the resistor (the area under the I–t graph is unchanged).

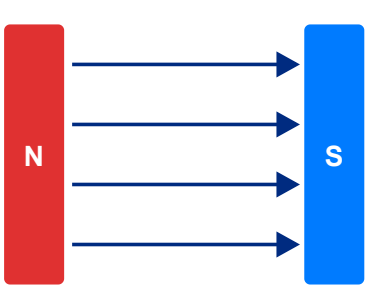
A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Magnetic Fields & Induction

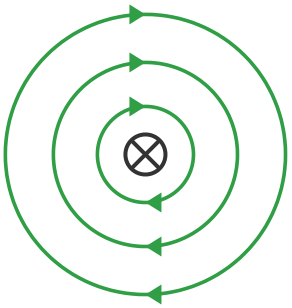
A magnetic field pushes sideways on moving charges, and that one fact explains electric motors, the curved tracks in particle detectors and the beam steering in old TV tubes. Run it backwards and you get electromagnetic induction: move a magnet near a coil and a voltage appears from nowhere. That is how almost all of the world's electricity is generated. Let's build both ideas up from the force on a single wire.

Magnetic fields and flux density

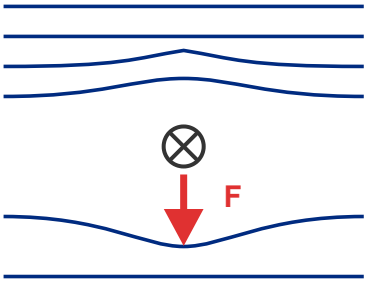
A **magnetic field** is a region where a magnet, a current-carrying wire or a moving charge feels a force. We draw it with field lines that run from north to south outside a magnet. The closer together the lines, the stronger the field. Between two flat poles the lines are straight, parallel and evenly spaced: a **uniform** field. A current in a straight wire makes circular field lines around it, in the direction given by the **right-hand grip rule** (thumb along the current, fingers curl the way the field goes).



uniform field between poles
(B the same everywhere)



wire, current into the page
(right-hand grip: clockwise)



lines bunch above, thin out below:
the wire is pushed down

Above the wire its own field points the same way as the magnet's field and they add; below, they point opposite ways and partly cancel. The wire is pushed from the strong-field side towards the weak-field side.

The strength of a magnetic field is measured by its **magnetic flux density** B. It is defined by the force it exerts on a wire carrying a current at right angles to the field: the bigger the force on each metre of wire for each amp of current, the stronger the field.

DEFINITION

Magnetic flux density B is the force per unit length per unit current on a conductor placed at right angles to the field, $B = F/Il$. Its unit is the **tesla**: 1 T is the flux density that exerts a force of 1 N on a 1 m length of

wire carrying 1 A at right angles to the field ($1\ \mathrm{T} = 1\ \mathrm{N\ A^{-1}\ m^{-1}}$). B is a vector, in the direction of the field lines.

For scale: the Earth's field is about 5×10^{-5} T, a fridge magnet about 0.01 T, a strong laboratory magnet about 1 T and an MRI scanner 1.5 to 3 T.

Force on a current-carrying wire: F = BIl sin θ

The force on a wire in a magnetic field is proportional to the flux density, the current and the length of wire in the field. It also depends on the angle θ between the wire and the field. Only the part of the current that cuts **across** the field lines feels a force, so a wire lying along the field feels nothing at all.

Force on a current-carrying conductor

FORMULA SHEET

$$F = BIl\sin\theta$$

F: force / N B: flux density / T I: current / A l: length in the field / m

θ: angle between wire and field

The force is always at right angles to **both** the field and the current. To find which way, use **Fleming's left-hand rule**: hold your thumb, first finger and second finger of your left hand at right angles to one another.

- **F**irst finger: **F**ield (north to south)
- se**C**ond finger: **C**urrent (conventional, + to −)
- thu**M**b: **M**otion, the direction of the force

EXAM TIP

In Fleming's rule the current finger must be at right angles to the field finger. If the wire is at an angle, use the component of the current at right angles to the field. That is exactly what the $\sin\theta$ in the equation does.

WORKED EXAMPLE

A straight wire carries a current of 2.5 A. A 6.0 cm length of it lies in a uniform magnetic field of flux density 0.15 T, at an angle of 40° to the field. Calculate the force on the wire. What is the largest force this wire could feel in this field, and how would you arrange it?

Show solution

$F = BIl\sin\theta = 0.15 \times 2.5 \times 0.060 \times \sin 40^\circ = 0.0225 \times 0.643 = 1.4 \times 10^{-2}$ N

The largest force is when $\sin\theta = 1$, with the wire at 90° to the field: $F = 0.15 \times 2.5 \times 0.060 = 2.3 \times 10^{-2}$ N. The direction is perpendicular to both the wire and the field, from Fleming's left-hand rule.

Force on a moving charge: F = Bqv sin θ

A current is just moving charges, so the force on a wire must really be the sum of forces on all the charge carriers inside it. A single charge moving through a field feels a force too, even with no wire at all.

From F = BIl to F = Bqv

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Take a wire of cross-sectional area A with n charge carriers per unit volume, each of charge q, drifting at speed v. From Unit 2, the current is:

$$I = nqvA$$

2. With the wire at right angles to the field, the force on a length l is:

$$F = BIl = B(nqvA)\,l$$

3. The number of charge carriers in that length of wire is (number per unit volume) × (volume):

$$N = nAl$$

4. Divide the total force by the number of carriers to get the force on each one:

$$\frac{F}{N} = \frac{B\,nqvA\,l}{nAl} \quad \Rightarrow \quad \boxed{F = Bqv}$$

If the charge moves at an angle θ to the field, only the velocity component $v\sin\theta$ across the field counts, giving $F = Bqv\sin\theta$.

Force on a moving charge

FORMULA SHEET

$$F = Bqv\sin\theta$$

q: charge / C v: speed / m s^{-1} θ: angle between velocity and field

Fleming's left-hand rule still works, but the second finger points along the **conventional current**. For a positive charge that is the way it moves. For an electron (or any negative charge) the current points **opposite** to its velocity, so the force is reversed.

Circular motion in a uniform field

The magnetic force is always at right angles to the velocity. That has two big consequences:

- It does **no work** on the particle (force × distance moved in the direction of the force is zero), so the particle's speed and kinetic energy never change.
- It changes the particle's direction at a steady rate, so in a uniform field at right angles to its motion the particle moves in a **circle**. The magnetic force provides the centripetal force.

Setting the magnetic force equal to the centripetal force, $Bqv = mv^2/r$, gives the radius of the circle, $r = mv/BQ = p/BQ$. That result is **derived** in full on the Nuclear & Particle Physics page, where it is used to measure the momentum of particles from their curved tracks.

WORKED EXAMPLE

An electron moves at 3.0×10^7 m s^{-1} at right angles to a uniform magnetic field of flux density 4.0 mT. Calculate the magnetic force on it and its acceleration.

Show solution

$F = Bqv = 4.0 \times 10^{-3} \times 1.60 \times 10^{-19} \times 3.0 \times 10^7 = 1.9 \times 10^{-14}$ N

$a = F/m = 1.92 \times 10^{-14}/9.11 \times 10^{-31} = 2.1 \times 10^{16}$ m s^{-2}, directed towards the centre of its circular path. Tiny force, enormous acceleration, because the electron's mass is so small.

Magnetic flux and flux linkage

To talk about induction we need a way to say "how much field passes through a coil". Picture the field lines threading through a loop of wire. If the loop faces the field, many lines pass through; if it is tilted, fewer do; if it lies edge-on to the field, none do. **Magnetic flux** Φ measures this. For an area A at right angles to a uniform field, $\Phi = BA$. That is also why B is called flux **density**: it is flux per unit area.

Magnetic flux through an area

$$\Phi = BA\cos\theta$$

Φ: flux / Wb A: area / m² θ: angle between the field and the normal to the area

The $\cos\theta$ is there because only the component of B at right angles to the area threads through it. When the area faces the field squarely, $\theta = 0$ and $\Phi = BA$.

A coil of N turns has the same flux through every turn, and each turn will contribute to the induced e.m.f., so we multiply: the **flux linkage** is $N\Phi$.

DEFINITION

Magnetic flux Φ through an area A at right angles to a uniform field B is $\Phi = BA$. Its unit is the **weber**, $1\ \text{Wb} = 1\ \text{T m}^2$.

Flux linkage is the flux through a coil multiplied by the number of turns, $N\Phi$. Its unit is also the weber (sometimes written Wb-turns).

Electromagnetic induction: a magnet and a coil

Push a magnet into a coil connected to a sensitive meter and the needle kicks. Hold the magnet still inside the coil and the needle returns to zero. Pull it out and the needle kicks the other way. An e.m.f. is **induced** in the coil, but only while the flux linkage through it is **changing**.

Only the **relative** motion matters. Seen from the magnet, it is the wire that moves through the field, so its free electrons feel the Bqv force and are pushed along the wire: the wire is "cutting" field lines. Stop the motion and the push stops. Whether you move the magnet or the coil, the e.m.f. is the same.

KEY IDEA

Factors affecting the e.m.f. induced by a moving magnet

The size of the induced e.m.f. is **bigger** if you:

- move the magnet (or the coil) **faster**, so the flux linkage changes more quickly;
- use a **stronger magnet**, so the change in flux is bigger;
- use a coil with **more turns**, so the flux links more turns;
- use a coil with a **larger cross-sectional area** (more flux through each turn).

The **direction** of the e.m.f. reverses if you reverse the motion (in instead of out) or turn the magnet round (south pole first). There is no e.m.f. at all when there is no relative motion.

COMMON MISTAKE

The induced **e.m.f.** does not depend on the resistance of the circuit, but the induced **current** does ($I = \varepsilon/R$). If the coil is not part of a complete circuit there is still an e.m.f. across its ends, but no current flows.

Faraday's law and Lenz's law

The simulation above shows the pattern: the e.m.f. is large where the flux linkage graph is steep and zero where it is flat. The e.m.f. is the **rate of change of flux linkage**. That is **Faraday's law**. The direction comes from **Lenz's law**, and the two combine into one equation.

DEFINITION

Faraday's law: the magnitude of the induced e.m.f. is equal to the rate of change of flux linkage.

Lenz's law: the direction of the induced e.m.f. (and of any induced current) is such as to oppose the change that produces it.

Faraday's and Lenz's laws

FORMULA SHEET

$$\varepsilon = -\frac{d(N\Phi)}{dt}$$

ε: induced e.m.f. / V $N\Phi$: flux linkage / Wb t: time / s

$d(N\Phi)/dt$ is the gradient of a flux linkage–time graph. If the flux linkage changes steadily, the e.m.f. is just $\Delta(N\Phi)/\Delta t$. If it does not, that ratio gives the **average** e.m.f., and the instantaneous e.m.f. is the gradient of the tangent. The minus sign is Lenz's law: the e.m.f. acts to **oppose** the change in flux linkage. In calculations of the size of the e.m.f. you can leave it out.

KEY IDEA

Why Lenz's law has to be true: energy conservation

Suppose the induced current helped the change instead of opposing it. Push a magnet towards a coil; the induced current would attract the magnet further in, so it would speed up, which would induce a bigger

current, which would pull harder... The magnet would accelerate and the coil would heat up, all from one small push. That would create energy from nothing.

So the induced current must **oppose** the motion. You have to do work pushing the magnet against the repulsion, and that work is exactly the electrical energy that appears in the circuit. Lenz's law is the conservation of energy applied to induction.

A rod moving through a field

The cleanest case of all is a straight conductor moving at right angles to a uniform field, like an aircraft wing or a metal rod sliding along rails.

The e.m.f. across a moving rod, ε = Blv

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A rod of length l moves at speed v at right angles to a uniform field B. In a time Δt it moves a distance $v\,\Delta t$, sweeping out an area:

$$\Delta A = l\,v\,\Delta t$$

2. The flux it cuts in that time (one "turn", so $N = 1$) is:

$$\Delta\Phi = B\,\Delta A = Blv\,\Delta t$$

3. Faraday's law: the e.m.f. is the rate of change of flux linkage:

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{Blv\,\Delta t}{\Delta t} \quad \Rightarrow \quad \boxed{\varepsilon = Blv}$$

4. Check it with forces instead. Each free electron in the rod moves with it, so feels $F = Bev$ along the rod. Electrons pile up at one end until the electric field this creates balances the magnetic push, $eE = Bev$. With $E = V/l$:

$$e\frac{V}{l} = Bev \quad \Rightarrow \quad V = Blv$$

Same answer: "cutting flux" and "the force on moving charges" are two descriptions of the same physics.

WORKED EXAMPLE

An aircraft with a wingspan of 35 m flies horizontally at 240 m s^{-1} where the vertical component of the Earth's magnetic field is 4.5×10^{-5} T. Calculate the e.m.f. induced between its wingtips. Explain why this e.m.f. could not be used to light a lamp in the cabin.

Show solution

The wings cut the vertical component of the field: $\varepsilon = Blv = 4.5 \times 10^{-5} \times 35 \times 240 = 0.38$ V

Any wires connecting the wingtips to a lamp travel with the aircraft through the same field, so the same e.m.f. is induced in them, in the same direction round the loop. The two e.m.f.s cancel, the flux linkage of the complete circuit does not change, and no current flows.

A coil rotating in a field

Rotate a flat coil in a uniform field and the angle between the field and the coil keeps changing, so the flux linkage rises and falls. Here you can see Faraday's law as a graph: the e.m.f. at each instant is minus the gradient of the flux linkage graph.

COMMON MISTAKE

A very common error is to say the e.m.f. is biggest when the flux linkage is biggest. It is the other way round for a rotating coil: the e.m.f. depends on how fast the flux linkage is **changing**, not on how big it is.

WORKED EXAMPLE

A coil of 250 turns and area 3.0 cm² lies with its plane at right angles to a uniform field of 0.40 T. The field is switched off and falls steadily to zero in 20 ms. (a) Calculate the e.m.f. induced. (b) The coil is part of a circuit of total resistance 5.0 Ω. Calculate the current and the total charge that flows.

Show solution

(a) Initial flux linkage: $N\Phi = NBA = 250 \times 0.40 \times 3.0 \times 10^{-4} = 3.0 \times 10^{-2}$ Wb. It falls to zero, so $\Delta(N\Phi) = 3.0 \times 10^{-2}$ Wb.

$$\varepsilon = \frac{\Delta(N\Phi)}{\Delta t} = \frac{3.0 \times 10^{-2}}{20 \times 10^{-3}} = 1.5 \text{ V}$$

(b) $I = \varepsilon / R = 1.5 / 5.0 = 0.30$ A, flowing for 20 ms, so $Q = It = 0.30 \times 0.020 = 6.0 \times 10^{-3}$ C.

Notice that $Q = \Delta(N\Phi)/R$: switching the field off twice as fast would double the current but halve the time, so the same charge flows.

Mutual induction: two linked coils

You don't need a moving magnet to change the flux through a coil. Put two coils side by side. A current in the first (the **primary**) makes a magnetic field, and some of that field threads through the second (the **secondary**). Change the current in the primary and the flux linkage of the secondary changes, so an e.m.f. is induced in it, even though nothing moves and the coils are not electrically connected.

KEY IDEA

Factors affecting the e.m.f. induced by a changing current in a linked coil

- The **rate of change of the current** in the primary: switching on or off quickly, or changing the current quickly with a rheostat, gives a bigger e.m.f. A steady current, however large, gives none.
- The **size of the change** in current: a bigger current makes a bigger change in flux when it is switched.
- The **number of turns** on the secondary (and on the primary, which sets how much field each amp makes).
- How much of the primary's flux **links** the secondary: putting the coils closer and on the same axis, or winding both on a shared **soft iron core**, which concentrates the flux and carries it from one coil to the other.

By Lenz's law, while the primary current is **growing** the secondary's induced current makes a field that opposes the primary's field. While it is **falling**, the secondary's field points the same way as the primary's, trying to keep the flux up.

EXAM TIP

Exam questions often show a primary current–time graph and ask you to sketch the secondary e.m.f. Use $\varepsilon \propto -\text{gradient}$: wherever the current graph is flat (constant current, even if large), the e.m.f. is zero; where it is a straight sloping line, the e.m.f. is a constant; steeper slopes give bigger e.m.f.s; and the sign flips when the slope changes from up to down.

WORKED EXAMPLE

The flux linkage of a secondary coil rises steadily from 0 to 0.090 Wb in 5.0 ms when the primary is switched on, stays constant for 20 ms, then falls steadily back to zero in 2.0 ms when the primary is switched off. Calculate the e.m.f. in each stage and sketch the e.m.f.–time graph.

Show solution

Rising: $\varepsilon = -\dfrac{0.090}{5.0 \times 10^{-3}} = -18\ \text{V}$ for 5.0 ms.

Constant: the gradient is zero, so $\varepsilon = 0$ for 20 ms.

Falling: $\varepsilon = -\dfrac{-0.090}{2.0 \times 10^{-3}} = +45\ \text{V}$ for 2.0 ms.

The graph is a rectangle of −18 V lasting 5 ms, a gap at zero, then a taller, narrower rectangle of +45 V lasting 2 ms. Both rectangles have the same area (0.090 V s = 0.090 Wb), because the flux linkage returns to where it started.

Quick checks

QUICK CHECK

Which of these is equivalent to the tesla?

- N A m
- kg m s⁻² A⁻¹
- kg s⁻² A⁻¹
- Wb m²

From $B = F/Il$: $\text{T} = \text{N A}^{-1}\text{m}^{-1} = (\text{kg m s}^{-2})\,\text{A}^{-1}\text{m}^{-1} = \text{kg s}^{-2}\text{A}^{-1}$. Also $\text{T} = \text{Wb m}^{-2}$, not Wb m².

QUICK CHECK

An electron moves from left to right across the page through a uniform magnetic field directed into the page. Which way is the magnetic force on it?

- Towards the top of the page
- Towards the bottom of the page
- Into the page
- There is no force, because the electron moves at right angles to the field

The conventional current is opposite to the electron's motion, so it points right to left. First finger into the page, second finger to the left: the thumb points down the page. (A proton moving the same way would be pushed up.)

QUICK CHECK

A proton moves in a circle in a uniform magnetic field. How much work does the magnetic force do on it in one complete orbit?

- None, because the force is always perpendicular to the velocity
- Bqv × 2πr
- None, because the proton returns to its starting point, but some is done during each half orbit
- It depends on the speed of the proton

Work = force × displacement in the direction of the force. The force always points to the centre while the motion is along the circle, so no work is done at any instant. That is why the speed stays constant.

QUICK CHECK

A flat coil rotates steadily in a uniform magnetic field. At which position is the induced e.m.f. largest?

- When the plane of the coil is at right angles to the field, so the flux linkage is maximum
- When the plane of the coil is parallel to the field, so the flux linkage is zero
- When the coil is at 45° to the field
- The e.m.f. is the same at every position

The e.m.f. is the rate of change of flux linkage, the gradient of the $N\Phi$–t graph. That is steepest where $N\Phi$ passes through zero, and flat at the maximum.

QUICK CHECK

A bar magnet is dropped vertically through a coil connected to a data logger. Which describes the e.m.f.–time graph?

- Two identical pulses in the same direction
- Two pulses in opposite directions, the second smaller and wider
- Two pulses in opposite directions, the second taller and narrower
- A single pulse while the magnet is inside the coil

The flux linkage rises as the magnet enters and falls as it leaves, so the e.m.f.s are opposite. The magnet accelerates, so it leaves faster than it enters: the flux changes faster (taller pulse) for a shorter time (narrower). The areas are equal because the total change in flux linkage is the same.

QUICK CHECK

Which change would not increase the e.m.f. induced when a magnet is pushed into a coil?

- Using a coil with more turns
- Pushing the magnet in faster
- Using a stronger magnet
- Using a coil made of wire with a higher resistance

The e.m.f. depends only on the rate of change of flux linkage. A higher resistance would reduce the induced **current**, but not the e.m.f.

A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Nuclear & Particle Physics

What is everything made of? A century ago the answer was "atoms"; then it was "electrons, protons and neutrons"; now it is quarks and leptons. Each step came from firing particles at other particles, harder and harder, and reading the tracks that came out. This page follows that story: alpha scattering, electron beams, accelerators, detectors, and the conservation laws that decide which particle reactions can happen.

Inside the atom: nucleon and proton numbers

An atom has a tiny, dense, positively charged **nucleus** made of protons and neutrons (together called **nucleons**), surrounded by electrons. The nucleus is about 10^{-15} to 10^{-14} m across, while the atom is about 10^{-10} m across: if the nucleus were a marble, the atom would be the size of a football stadium. Yet the nucleus holds more than 99.9% of the mass.

A nucleus is labelled with two numbers written in front of the chemical symbol, ${}^{A}_{Z}\mathrm{X}$:

DEFINITION

The **nucleon number** (mass number) A is the total number of protons and neutrons in the nucleus.

The **proton number** (atomic number) Z is the number of protons in the nucleus. It decides which element the atom is. The number of neutrons is $A - Z$.

For example, ${}^{14}_{6}\mathrm{C}$ has 6 protons and $14 - 6 = 8$ neutrons. Nuclei of the same element with different numbers of neutrons, such as ${}^{12}_{6}\mathrm{C}$ and ${}^{14}_{6}\mathrm{C}$, are **isotopes**: same chemistry, different mass. An alpha particle is a helium nucleus, ${}^{4}_{2}\mathrm{He}$: 2 protons and 2 neutrons, charge $+2e$.

Alpha scattering and the nuclear model

How the picture of the atom changed

- **Early 1800s:** Dalton pictures atoms as tiny, solid, indivisible spheres.
- **1897:** Thomson discovers the electron, so atoms must have smaller parts. His "plum pudding" model has electrons dotted through a sphere of spread-out positive charge.
- **1909–1911:** Geiger and Marsden fire alpha particles at thin gold foil; Rutherford explains the results with a tiny positive **nucleus**.
- **1913:** Bohr puts the electrons in fixed energy levels around the nucleus.
- **1919 and 1932:** the proton is identified, then Chadwick discovers the neutron.

- **1960s onwards:** high-energy electron scattering shows protons and neutrons are themselves made of **quarks**.

Each model was replaced because an experiment produced results it could not explain. That is how scientific models change over time.

The experiment

A beam of alpha particles from a radioactive source was aimed at a very thin gold foil in a vacuum (so the alphas were not stopped by air). A movable detector, a zinc sulfide screen that gives a flash of light when hit, counted the alphas scattered at each angle. The results:

- **Almost all** the alpha particles passed straight through or were deflected by only a few degrees.
- A **small fraction** were deflected through large angles.
- A **very few** (roughly 1 in 8000) bounced back through more than 90°.

KEY IDEA

What the results tell us

- **Most alphas pass straight through**, so most of the atom is **empty space**.
- **A few are deflected through large angles**, so there must be a very strong electric field somewhere. That needs a large charge concentrated in a **very small volume**: the nucleus.
- **Some bounce straight back.** An alpha particle is about 7000 times more massive than an electron, so only something much more massive than the alpha could turn it round. Nearly all the **mass** must be in the nucleus too.
- The alphas are **repelled**, so the nucleus is **positively charged**.

In the plum pudding model the positive charge is spread over the whole atom, so its field is weak everywhere and the electrons are far too light to turn an alpha round. It predicts only tiny deflections, which is why it had to go.

WORKED EXAMPLE

An alpha particle with kinetic energy 5.0 MeV is fired directly at a gold nucleus (Z = 79). Calculate its distance of closest approach. What does this tell you about the size of the gold nucleus?

Show solution

At the closest point the alpha stops for an instant, so all its kinetic energy has become electric potential energy: $E_k = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r}$.

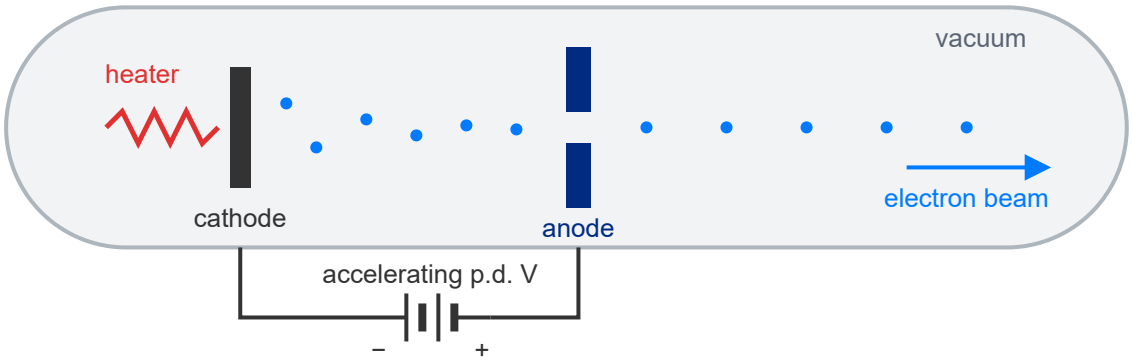
$E_k = 5.0 \times 1.60 \times 10^{-13} = 8.0 \times 10^{-13}$ J, $Q_1 = 79e$, $Q_2 = 2e$:

$$r = \frac{8.99 \times 10^{9} \times (79 \times 1.60 \times 10^{-19}) \times (2 \times 1.60 \times 10^{-19})}{8.0 \times 10^{-13}} = 4.5 \times 10^{-14}\ \text{m}$$

The alpha never touched the nucleus, so the radius of a gold nucleus must be **less than** about 4.5×10^{-14} m. (It is actually about 7×10^{-15} m.)

Electron beams: thermionic emission

To probe matter with electrons you first need a beam of them. Heat a metal filament (or a cathode next to it) to a high temperature and some of its free electrons gain enough energy to escape from the surface. This is **thermionic emission**. In a vacuum, the electrons are then accelerated towards a positive **anode**; some pass through a hole in it and form a narrow beam. The whole device is an **electron gun**.



An electron gun. The anode is positive relative to the cathode, so the electric field between them accelerates the emitted electrons. Beyond the anode there is no field, so they travel at constant velocity.

Each electron is accelerated through a p.d. V, so the electric field does work eV on it, which becomes kinetic energy. An electron accelerated through 1 V gains 1 electronvolt, $1\ \text{eV} = 1.60 \times 10^{-19}$ J.

$$eV = \tfrac{1}{2}mv^2 \quad \Rightarrow \quad v = \sqrt{\frac{2eV}{m}}$$

Once it has a beam, an **electric field** can deflect it (the force eE acts opposite to the field lines for an electron, so the beam bends in a parabola, like a projectile), and a **magnetic field** can bend it into a circular arc (the force Bev is always at right angles to the motion). Both ideas appear again in accelerators and detectors.

WORKED EXAMPLE

Electrons are emitted from a hot cathode and accelerated through a p.d. of 2.0 kV. Calculate their kinetic energy in joules and their speed.

Show solution

$E_k = eV = 1.60 \times 10^{-19} \times 2.0 \times 10^{3} = 3.2 \times 10^{-16}$ J (that is 2.0 keV)

$v = \sqrt{\dfrac{2E_k}{m}} = \sqrt{\dfrac{2 \times 3.2 \times 10^{-16}}{9.11 \times 10^{-31}}} = 2.7 \times 10^{7}\ \text{m s}^{-1}$, about 9% of the speed of light. At much higher p.d.s this simple equation stops working because the speed cannot exceed c.

Particle accelerators: linac and cyclotron

To smash particles together hard enough to see inside nucleons or create new particles, they need energies of GeV or more. A single accelerating gap would need billions of volts, which is impossible, so accelerators give the particle many smaller pushes. **Electric fields** do the accelerating; **magnetic fields** steer.

The linear accelerator (linac)

A linac is a long line of metal **drift tubes** in a vacuum, connected alternately to the two terminals of a high-frequency alternating supply. Inside a tube there is no electric field (the metal shields it), so the particle drifts at constant speed. In each **gap** between tubes there is a strong field, and the supply is timed so that the field always points forwards when the particle is crossing a gap. While the particle is hidden inside a tube, the supply reverses, ready for the next gap. Each gap adds energy QV.

The particle gets faster at every gap, but it must spend the same time (half a cycle of the supply) inside each tube. So the tubes must get **longer** along the accelerator.

The cyclotron

A cyclotron curls the path up so the same gap can be used again and again. Two hollow D-shaped electrodes, the **dees**, sit in a vacuum between the poles of a large magnet, with a gap between them and an alternating p.d. across the gap. Particles start near the centre.

- Inside a dee there is no electric field. The **magnetic field** makes the particle move in a **semicircle** (the magnetic force does no work, so the speed is constant).
- Each time the particle reaches the gap, the **electric field** across it accelerates the particle into the other dee. The p.d. has reversed in the meantime, so it pushes forwards both ways.
- Faster particles move in bigger circles ($r = p/BQ$), so the path spirals outwards until the particle reaches the edge and is extracted.

Why the cyclotron frequency doesn't depend on speed

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Inside a dee the particle moves in a semicircle of radius r at speed v. The time to go half way round is:

$$t_{1/2} = \frac{\pi r}{v}$$

2. The radius is $r = mv/BQ$ (derived in the next section). Substitute:

$$t_{1/2} = \frac{\pi}{v} \cdot \frac{mv}{BQ} = \frac{\pi m}{BQ}$$

The speed cancels. A faster particle goes round a bigger circle, and the two effects exactly balance.

3. One cycle of the supply must last two half-turns, so the period is $2\pi m/BQ$ and the frequency is:

$$\boxed{f = \frac{BQ}{2\pi m}}$$

So a fixed-frequency supply stays in step with the particle on every turn, from the centre to the edge.

WORKED EXAMPLE

A cyclotron for protons has dees of radius 0.50 m in a magnetic field of 1.5 T. Calculate (a) the frequency of the alternating supply and (b) the maximum kinetic energy of the protons, in MeV.

Show solution

(a) $f = \dfrac{BQ}{2\pi m} = \dfrac{1.5 \times 1.60 \times 10^{-19}}{2\pi \times 1.67 \times 10^{-27}} = 2.3 \times 10^{7}$ Hz (23 MHz)

(b) The largest orbit has $r = 0.50$ m, so the maximum momentum is $p = BQr = 1.5 \times 1.60 \times 10^{-19} \times 0.50 = 1.2 \times 10^{-19}$ kg m s^{-1}.

$$E_k = \frac{p^2}{2m} = \frac{(1.2 \times 10^{-19})^2}{2 \times 1.67 \times 10^{-27}} = 4.3 \times 10^{-12}\ \text{J} = \frac{4.31 \times 10^{-12}}{1.60 \times 10^{-13}}\ \text{MeV} = 27\ \text{MeV}$$

Detecting particles and r = p/BQ

Particles are far too small to see, so detectors rely on two effects.

- **Ionisation.** A fast charged particle knocks electrons out of the atoms it passes, leaving a trail of ions. In a cloud chamber the ions trigger droplets, in a bubble chamber tiny bubbles, and in modern detectors an electrical signal. Uncharged particles (neutrons, neutrinos, photons) hardly ionise, so they leave **no track**: you spot them from gaps and from where new tracks suddenly begin.
- **Deflection.** The detector sits in a magnetic field, so charged tracks curve. The **direction** of curving gives the sign of the charge, and the **radius** gives the momentum.

Deriving r = p/BQ

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. A particle of charge Q moves at speed v at right angles to a uniform field B. The magnetic force on it is:

$$F = BQv$$

This force is always perpendicular to the velocity, so it provides the centripetal force and the particle moves in a circle.

2. Set the magnetic force equal to the centripetal force:

$$BQv = \frac{mv^2}{r}$$

3. Cancel one factor of v and rearrange for r:

$$r = \frac{mv}{BQ}$$

4. mv is the momentum p, so:

$$\boxed{r = \frac{p}{BQ}}$$

A larger momentum gives a larger, less curved track; a stronger field or a larger charge curls it up more tightly.

Charged particle in a magnetic field

FORMULA SHEET

$$r = \frac{p}{BQ}$$

r: radius of path / m p: momentum / kg m s^{-1} B: flux density / T Q: charge / C

KEY IDEA

Reading particle tracks

- **Which way it curves** tells you the sign of the charge (use Fleming's left-hand rule, remembering that the current is opposite to the motion for a negative particle). Opposite charges curve opposite ways.
- **How tightly it curves** tells you the momentum: $p = BQr$. A spiral means the particle is losing momentum as it goes.
- **Thick, dense tracks** come from slow or heavily charged particles, which ionise strongly; fast light particles leave faint tracks.
- **Where tracks start or stop** without anything visible, an uncharged particle has arrived or left. Conservation of charge and momentum at that point tells you about the invisible particle.

WORKED EXAMPLE

A proton leaves a circular track of radius 0.25 m in a bubble chamber with a magnetic field of 1.2 T. Calculate its momentum, in SI units and in MeV/c, and its kinetic energy in MeV.

Show solution

$p = BQr = 1.2 \times 1.60 \times 10^{-19} \times 0.25 = 4.8 \times 10^{-20}\ \text{kg m s}^{-1}$

In MeV/c: $pc = 4.8 \times 10^{-20} \times 3.00 \times 10^{8} = 1.44 \times 10^{-11}\ \text{J} = 90\ \text{MeV}$, so $p = 90\ \text{MeV}/c$.

$E_k = \dfrac{p^2}{2m} = \dfrac{(4.8 \times 10^{-20})^2}{2 \times 1.67 \times 10^{-27}} = 6.9 \times 10^{-13}\ \text{J} = 4.3\ \text{MeV}$. Much less than its rest energy (938 MeV), so the non-relativistic formula is fine.

High energies, E = c²Δm and particle units

Why high energies are needed

There are two reasons. First, to **see** small detail you need a probe whose de Broglie wavelength $\lambda = h/p$ is about as small as the detail. A nucleon is about 10^{-15} m across, so the probe needs $p \approx h/\lambda = 6.6 \times 10^{-19}\ \text{kg m s}^{-1}$, which for an electron means an energy of about $pc \approx 2 \times 10^{-10}\ \text{J}$, roughly 1 GeV. Lower energies just diffract round the nucleon without resolving its insides.

Second, to **create** new particles you need enough energy to make their mass. Energy and mass are equivalent:

Mass–energy FORMULA SHEET

$$\Delta E = c^2 \Delta m$$

ΔE: energy / J Δm: mass / kg $c = 3.00 \times 10^8\ \text{m s}^{-1}$

In a high-energy collision, kinetic energy can turn into the mass of brand new particles. To discover a very massive particle, like the top quark, you need a collision with at least that much energy to spare.

Creation and annihilation

Energy can only become matter in **particle–antiparticle pairs**, because charge and the other quantum numbers must be conserved. In **pair production** a high-energy photon passing close to a nucleus becomes an electron and a positron. The photon needs at least enough energy for both rest masses, $2m_e c^2 = 1.02$ MeV; any extra becomes their kinetic energy. The nearby nucleus is needed to take up some momentum.

The reverse is **annihilation**: a particle meets its antiparticle and both vanish, their total energy reappearing as photons. An electron and positron at rest produce **two** photons of 0.511 MeV each, going in opposite directions so that the total momentum stays zero. PET scanners in hospitals detect exactly these photon pairs.

MeV, GeV and MeV/c²

Joules are inconveniently large for single particles, so particle physicists use electronvolts. Because $E = mc^2$, a mass can be given as an energy divided by c^2, and momentum as an energy divided by c.

Unit	Measures	In SI units
1 MeV	energy	$1.60 \times 10^{-13}\ \text{J}$
1 GeV	energy	$1.60 \times 10^{-10}\ \text{J}$
1 MeV/c²	mass	$\dfrac{1.60 \times 10^{-13}}{(3.00 \times 10^8)^2} = 1.78 \times 10^{-30}\ \text{kg}$
1 GeV/c²	mass	$1.78 \times 10^{-27}\ \text{kg}$
1 MeV/c	momentum	$5.33 \times 10^{-22}\ \text{kg m s}^{-1}$

Useful values: electron 0.511 MeV/c², proton 938 MeV/c², neutron 940 MeV/c². So a proton's rest mass is just under 1 GeV/c².

WORKED EXAMPLE

(a) Calculate the minimum photon energy, in MeV and in J, needed to produce an electron–positron pair, and the wavelength of such a photon. (b) A proton and an antiproton, both at rest, annihilate into two identical photons. Calculate the energy of each photon in GeV.

Show solution

(a) $E = 2m_e c^2 = 2 \times 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-13}\ \text{J} = 1.02\ \text{MeV}$.

$\lambda = \dfrac{hc}{E} = \dfrac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{1.64 \times 10^{-13}} = 1.2 \times 10^{-12}\ \text{m}$, a gamma ray.

(b) The total rest energy is $2 \times 938\ \text{MeV} = 1876\ \text{MeV}$, shared between two photons, so each has $938\ \text{MeV} = 0.938\ \text{GeV}$.

Longer lifetimes at high speed

Many particles are unstable and decay after a characteristic lifetime. When they travel at a speed close to the speed of light, their lifetime measured in the laboratory is **longer** than when they are at rest, so they travel much further before decaying. This relativistic effect is negligible at everyday speeds and only becomes significant when the speed is a large fraction of c. Two examples: muons made by cosmic rays 10–15 km up in the atmosphere reach the ground, although at rest they live only 2.2 μs (about 660 m of travel even at the speed of light); and unstable particles made in accelerators leave longer tracks than you would expect from their rest lifetimes.

The standard model: quarks and leptons

The hundreds of particles found in accelerators fit a simple pattern. All matter particles are either **quarks** or **leptons**, which as far as we know are **fundamental** (not made of anything smaller).

Quarks	Charge / e	Baryon number	Leptons	Charge / e	Lepton number
up u, charm c, top t	$+\frac{2}{3}$	$+\frac{1}{3}$	electron e^-, muon μ^-, tau τ^-	−1	+1
down d, strange s, bottom b	$-\frac{1}{3}$	$+\frac{1}{3}$	neutrinos ν_e, ν_μ, ν_τ	0	+1

Quarks are never found on their own. They combine into particles called **hadrons**, in only two ways:

- **Baryons** are made of **three quarks** (qqq), for example the proton (uud: $\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$) and the neutron (udd: $\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$). Every baryon has baryon number +1.
- **Mesons** are made of a **quark and an antiquark** (q $\bar{q}$), for example the pions: π^+ = u $\bar{d}$, π^- = d $\bar{u}$ and π^0 (a mixture of u $\bar{u}$ and d $\bar{d}$). Mesons have baryon number 0.

The other two classes are **leptons**, the electron and its relatives, which are fundamental and are not made of quarks; and **photons**, the particles of electromagnetic radiation.

Antiparticles

Every particle has an **antiparticle** with the **same mass** and the **opposite** charge, baryon number and lepton number. The positron e^+ has charge +1 and lepton number −1. The antiproton $\bar{p} = \bar{u}\,\bar{u}\,\bar{d}$ has charge −1 and baryon number −1. The antineutron $\bar{n} = \bar{u}\,\bar{d}\,\bar{d}$ is still uncharged, but it is not the same as a neutron, because its baryon number is −1. A few neutral particles, such as the photon and the π^0, are their own antiparticles.

A pattern that made a prediction

Quarks and leptons come in matching pairs, called generations: (u, d) with (e, ν_e); (c, s) with (μ, ν_μ); and (t, b) with (τ, ν_τ). By 1977 the tau lepton and the bottom quark had been found, but the bottom quark had no partner. The symmetry of the model demanded one, the **top quark**, with charge $+\frac{2}{3}$. It was so massive (about 173 GeV/c², nearly the mass of a gold atom) that it took until 1995, and a much more powerful accelerator, to make it. Its discovery was a big success for the model.

Conservation laws and particle equations

Particle reactions are written like chemical equations. In every reaction certain quantities must be the same before and after. If any one of them is not conserved, the reaction **cannot happen**.

KEY IDEA

Always conserved

- **Charge** Q
- **Baryon number** B: +1 for baryons, −1 for antibaryons, 0 for everything else (+⅓ per quark)
- **Lepton number** L: +1 for leptons, −1 for antileptons, 0 for everything else
- **Energy** (including rest energy) and **momentum**

To check a reaction, make a small table with a column for each particle and rows for Q, B and L, then compare the totals on each side.

For example, in beta-minus decay a neutron turns into a proton:

$$\mathrm{n} \rightarrow \mathrm{p} + \mathrm{e}^- + \bar{\nu}_e$$

	n	→	p	e^-	$\bar{\nu}_e$	Conserved?
Charge Q	0		+1	−1	0	0 = 0 ✓

Baryon number B	+1	+1	0	0	1 = 1 ✓
Lepton number L	0	0	+1	−1	0 = 0 ✓

Without the antineutrino, lepton number would go from 0 to +1, so the antineutrino **must** be there. (Its existence was first suggested because the electrons from beta decay had a spread of energies, which only made sense if an unseen particle was carrying some away.) At the quark level, one down quark has become an up quark: $\mathrm{d} \rightarrow \mathrm{u} + \mathrm{e}^{-} + \bar{\nu}_{e}$.

EXAM TIP

Lepton number catches most people out. Remember that an **anti**neutrino and a **positron** both have $L = -1$. And don't forget energy: a single particle at rest can only decay into particles whose total rest mass is **smaller** than its own. That is why a free neutron can decay into a proton, but a free proton (lighter than a neutron) cannot decay into a neutron.

WORKED EXAMPLE

A negative pion collides with a stationary proton: $\pi^{-} + \mathrm{p} \rightarrow \mathrm{n} + \mathrm{X}$. Use conservation laws to identify X. Then explain why the reaction $\mathrm{p} \rightarrow \mathrm{n} + \pi^{+}$ cannot happen for a free proton.

Show solution

Charge: $-1 + 1 = 0 + Q_X$, so $Q_X = 0$. Baryon number: $0 + 1 = 1 + B_X$, so $B_X = 0$. Lepton number: $0 = 0 + L_X$, so $L_X = 0$.

X is uncharged with $B = 0$ and $L = 0$: a neutral meson such as the π⁰ (or a photon).

For $\mathrm{p} \rightarrow \mathrm{n} + \pi^{+}$: charge (+1 = 0 + 1), baryon number (1 = 1 + 0) and lepton number (0 = 0) are all conserved. But the products have a total rest energy of $940 + 140 = 1080$ MeV, more than the proton's 938 MeV. A particle at rest cannot decay into heavier products, so energy conservation forbids it.

Quick checks

QUICK CHECK

In the alpha-scattering experiment, what was the evidence that the nucleus contains most of the atom's mass?

- Most alpha particles passed straight through the foil.
- A few alpha particles were scattered back through more than 90°.
- The alpha particles were repelled, not attracted.
- The foil had to be very thin.

Only something much more massive than the alpha particle could turn it right round. Passing straight through shows the atom is mostly empty space; being repelled shows the nucleus is positive.

QUICK CHECK

A proton and an alpha particle move at the same speed at right angles to the same uniform magnetic field. What is the ratio (radius of alpha's path) / (radius of proton's path)?

- 1/2
- 1
- 2
- 4

$r = mv/BQ$. The alpha has 4 times the mass but twice the charge, so its radius is $4/2 = 2$ times as big.

QUICK CHECK

Why do the drift tubes in a linear accelerator get longer along its length?

- To give a larger accelerating p.d. at each gap
- The particle speeds up, but must spend the same time (half a cycle) inside each tube
- Because the particle's mass increases, so it needs more room
- To reduce the magnetic field needed to steer the beam

The supply frequency is fixed, so the particle must arrive at each gap exactly half a cycle after the last one. Faster particles cover more distance in that time, so each tube must be longer.

QUICK CHECK

Which of these is a meson?

- u u d
- u $\bar{d}$
- $\bar{u}$ $\bar{u}$ $\bar{d}$
- u d

A meson is a quark plus an antiquark: u $\bar{d}$ is the π^+. uud is a proton (baryon), $\bar{u}$ $\bar{u}$ $\bar{d}$ an antiproton (antibaryon), and u d is not allowed (charge +⅓).

QUICK CHECK

Which property of the electron antineutrino $\bar{\nu}_e$ is correct?

- Charge −1, lepton number +1
- Charge 0, lepton number +1
- Charge 0, lepton number −1
- Charge +1, lepton number −1

An antiparticle has the opposite charge and lepton number of its particle. The neutrino has charge 0 and $L = +1$, so the antineutrino has charge 0 and $L = -1$.

QUICK CHECK

Which reaction is impossible because baryon number is not conserved?

- $\pi^+ \rightarrow \mu^+ + \nu_\mu$
- $p + p \rightarrow p + p + \pi^0$
- $p \rightarrow \pi^+ + \pi^0$
- $e^- + e^+ \rightarrow \gamma + \gamma$

The proton has $B = +1$ but pions are mesons with $B = 0$, so B would go from 1 to 0. Charge is conserved (+1 = +1 + 0), but one failure is enough.